

Modeling chromosome spatial organization with polymer self-consistent field theory

Integrating Hi-C contact maps and epigenomic data to infer affinity binding sites

by

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Abstract

Chromosomes are spatially organized into topologically associating domains (TADs) during interphase, a process mediated in part by Structural Maintenance of Chromosomes (SMC) complexes. The The Strings and Binders Switch (SBS) method has previously been demonstrated using a coarse-grained polymer model with effective binding sites that capture overall molecular interactions, and has been applied in molecular dynamics (MD) simulations to reconstruct three-dimensional chromosome structure. In this work, we model chromatin using polymer self-consistent field theory (SCFT). We implemented both SCFT and effective binding sites inferring codebases, in which interactions between polymer blocks are described by a mean-field Flory–Huggins model. We further incorporated epigenetic information by assigning A/B-compartment properties to polymer blocks and introducing compartment-dependent interaction parameters. Using SCFT, we obtained the polymer morphology, an overlap-based contact map, and the free-energy landscape for a 2-Mbp chromatin segment. We find that SCFT generates a contact map that is quantitatively comparable to experimental Hi-C data. In addition, we characterized the free-energy landscape of an ABC triblock copolymer benchmark system by analyzing its first-order phase transition. We evaluated how interaction enthalpy and conformational entropy vary with interaction strength in the chromatin model, and related these thermodynamic contributions to the similarity between simulated and experimental contact maps.

Introduction

2.1. Chromatin spatial organization

Eukaryotic genomes are organized into topologically associating domains (TADs). TADs are sub-megabase-scale chromatin regions within which genome loci interact with each other more frequently than with loci outside the domain. As shown in Figure 2.1a, TAD boundaries are associated with proteins that regulate the chromosome, including CCCTC-binding factor (CTCF) and structural maintenance of chromosome (SMC). These boundaries regulate interactions between enhancers and their target genes, thereby playing a crucial role in regulating gene expression.[1]

To capture TAD-scale genome organization in vivo, a series of chromosome conformation capture (3C) methods has been developed. The original 3C protocol quantifies contact frequency between a predefined pair of genomic loci [2]. Subsequent extensions, including 4C, 5C [3], and Hi-C, expanded this principle to many loci in parallel with much higher throughput. Hi-C, in particular, enables genome-wide contact mapping at kilobase-scale resolution [4].

The core steps of Hi-C are illustrated in Figure 2.1c. First, the DNA-protein-DNA interaction is stabilized by formaldehyde. The DNA is then digested with a restriction enzyme, and the resulting 5' overhangs are filled in to generate blunt ends while incorporating a biotin label. Proximity ligation joins DNA ends that were held near each other within cross-linked chromatin complexes, producing chimeric fragments that encode 3D proximity information. These ligation products are enriched using streptavidin beads and sequenced by paired-end sequencing to identify the paired genomic coordinates. The resulting Hi-C contact map represents an ensemble average over millions of cells.

In addition to sequencing-based assays, three-dimensional chromosome localization can be measured directly, demonstrating that TADs are also manifest in physical space. The most commonly used technology in the field is the fluorescence in situ hybridization (FISH). In the super-resolution FISH microscopy experiment performed by Bintu et al. [5], consecutive 30 kb segments in multiple 1.2-2.5 Mb regions were labeled and imaged. The region is labeled with primary oligo-paint probes, each containing a segment-specific readout sequence. Dye-labeled readout probes were then introduced iteratively for imaging, followed by bleaching or removal. From imaging hundreds of chromosomes, it was found that spatial overlap and median distance are strongly correlated with Hi-C contact frequency.

2.2. Single molecule mechanism of chromosome organization

Structural maintenance of chromosomes (SMC) complexes play vital roles at all stages of the cell cycle, including establishing interphase organization (e.g., TADs) and packaging chromatin into a highly ordered X-shaped structure during cell division. SMC complexes are multi-subunit motor proteins, as shown schematically in Figure 2.1b, which consume ATP to extrude DNA into loops. There are three classes of SMC complexes in eukaryotic cells: cohesin, condensin, and SMC5/6.

During interphase, the chromosome organization is mainly governed by cohesin. The cohesin binds

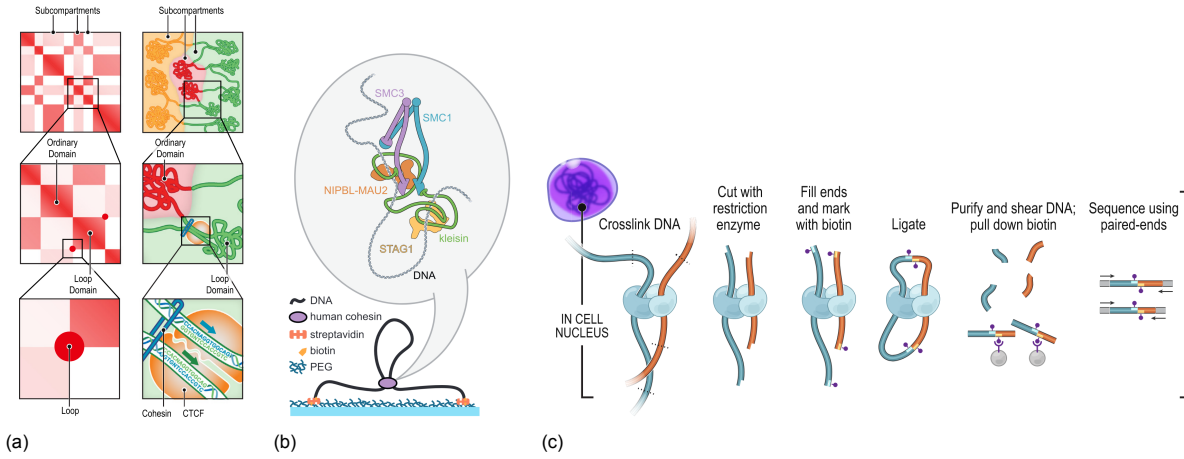


Figure 2.1: **(a)** Hi-C pattern illustrating TADs where chromatin is compacted into compartments (adapted from [4]). **(b)** Schematic of a DNA loop extrusion assay and the architecture of human cohesin (adapted from [6]). **(c)** Overview of the Hi-C experimental workflow (adapted from [4]).

to DNA and extrudes the loop asymmetrically, with the motor alternately switching the extrusion side. As a net effect, cohesin appears to localize along DNA in a symmetric, diffusive manner [6]. Upon encountering a CTCF molecule in convergent orientation, cohesin loop extrusion stalls at the CTCF-bound site, while continuing to reel in DNA from the other side, until it meets a CTCF molecule in the other orientation [7, 8]. CTCF is a DNA-binding protein that asymmetrically binds to specific non-palindrome sequence, so-called CTCF binding sites, which act as boundaries of TADs.[9]

During cell division, the roles of SMC complexes alter. Condensins become the primary loop extrusion motors, while cohesin maintains the sister chromatid. Condensin II first extrudes large DNA loops (~400 kb) in prophase. Then, when the nuclear envelope breaks down, condensin I, which is initially localized in the cytoplasm, binds to DNA and extrudes loops (~80 kb) nested within the large loops. SMC complexes also play a role in double-strand break (DSB) repair and homologous recombination. SMC5/6 and cohesin are recruited to DSB, and the loop extrusion by them may facilitate spreading of the DSB response marker, including histone phosphorylation.

SMC complexes extrude loops at a remarkably high rates. In an *in vitro* experiment performed by Ganji et al [10] (Figure 2.1b), yeast condensin showed an average extrusion speed of 0.6 kbp/s and up to 1.5 kbp/s, depending on the DNA tension. Comparable rates have been reported for other SMC complexes, for example, 0.5-2.1 kbp/s for human cohesin [11, 12], and ~1 and 0.5 kbp/s for human condensin I and II, respectively [13]. The force-dependent step size has been measured at ~20-45 nm, corresponding to a few hundred base pairs per step. The characteristic step length is comparable to the size of the SMC complex, suggesting that stepping involves a complex-wide conformational change [14].

Despite having a fast stepping rate, the SMC complexes' ATP hydrolysis rates are low, on the order of 1 or a few ATP molecules per second, implying a low looping force. The stalling force was measured to be around 0.1-1 pN [15]. Quantitatively, the hydrolysis energy of an ATP is of the order of $\sim 20 k_B T$. The stepping energy is around $5-17 pN \cdot nm$ (approximately $1-4 k_B T$). Expressed at the scale of a DNA kuhn length (~100 nm), this corresponds to an energy cost on the order of several $k_B T$, providing a useful reference scale for interpreting energetic terms in polymer models.

From an entropic perspective, the SMC complexes consume energy to drive chromatin into lower entropy, more ordered conformations. One advantage of the SCFT model is that it provides the entropy and free-energy landscape. We return to this connection between energy scale and polymer conformation in section 5.1.

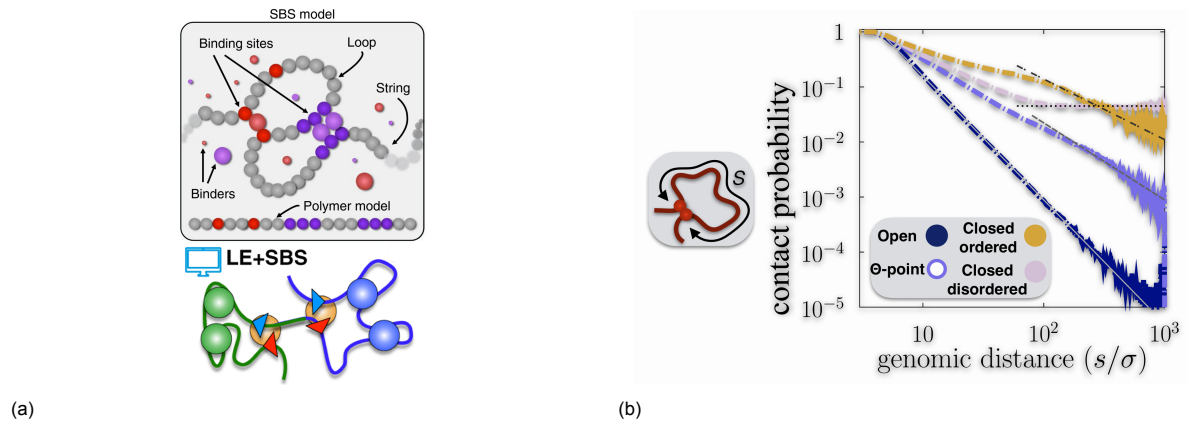


Figure 2.2: **(a)** Schematic of the SBS model, where chromatin is a bead–spring polymer containing discrete binding-site types (“colors”) that interact via diffusing cognate binders; and of the LE+SBS model, which additionally includes loop-extruding motors that form loops and can stall at CTCF-associated anchor sites (adapted from [17] and [19]). **(b)** Representative contact-probability scaling $P_c(s)$ across polymer regimes (coil, θ -point, globule), illustrating distinct power-law exponents and a plateau at large genomic separation (adapted from [16]).

2.3. Spring-bead and loop extrusion Model

The Strings and Binders Switch (SBS) polymer model [16, 17] is a well-established coarse-grained molecular dynamics (MD) simulation method for interpreting chromosomal 3D organization from Hi-C data. SBS is capable of generating an ensemble of conformations, whose contact and distance statistics are comparable to those from super-resolution microscopy [5]. The SBS model directly motivates this thesis; hence, we will discuss it in detail.

In the SBS model, the chromatin is represented as a coarse-grained, self-avoiding string of beads that carries discrete binding-site types (often described as bead “colors”). Diffusing cognate molecular binders from the nucleoplasm can bridge beads of the same type. The SBS model demonstrates that the chromatin polymer can exist in various conformations and phases, controlled by binder concentration and binding energy [16]. At low values, the chromatin is in coil conformation (open), whereas at high values, the chromatin phase-separates to be in globule conformation (closed). The conformation can be characterized by the contact probability $P_c(s)$ of bead pairs versus their genome contour distance s . As shown in Figure 2.2b, in the coil state, $P_c(s)$ decreases as a power law $P_c(s) \propto s^{-a}$, with an exponent $a \approx 2.1$. At the θ -point, where the switching between two states happens, the exponent $a \approx 1.5$. In the globule state, $P_c(s)$ first decrease with an exponent close to $a \approx 1.0$, then reach a long plateau at large separations.

To incorporate active loop formation by SMC complexes, the SBS model can be combined with the loop-extrusion (LE) model [18, 19]. In LE, extruding motor proteins load onto chromatin and extrude loops until they stall at CTCF sites in convergent orientation or collide with other motors. The combined LE+SBS model captures situations where active extrusion and binder-mediated phase separation act simultaneously, and it reproduces both TAD-like boundary features and pair median distance observed in Hi-C and microscopy.

A key step in the SBS or LE+SBS model is to infer the binding-site types (the bead “colors”) from the Hi-C contact map. The PRISMR approach [16, 17, 20] performs this inference via stochastic optimization, i.e., Monte Carlo and simulated annealing. It has been shown to be capable of analyzing higher-order chromatin structure from genome-wide Hi-C data. PRISMR has been used to predict how structural variants can reconfigure the 3D organization. Binding site patterns inferred by PRISMR show statistical correspondence with combinations of chromatin architectural factors, including CTCF. The details of this algorithm will be discussed in section 3, where we build our own code to recreate the features of PRISMR, due to the original PRISMR code is not publicly available.

In the MD implementation of the SBS/LE+SBS model, each bead represents a DNA segment of approximately 30 kb within a 2-Mb genome region. Beads are connected via FENE bonds (finitely extensible nonlinear elastic) springs to form the backbone. The Weeks-Chandler-Anderson (WCA)

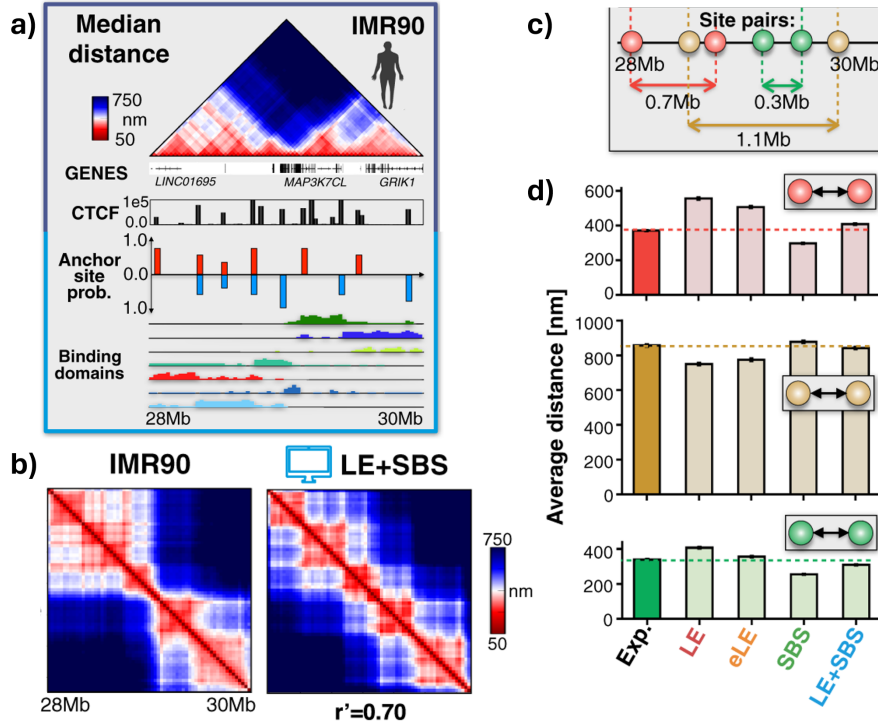


Figure 2.3: **(a)** Experimental and model inputs for a 2 Mb locus: microscopy median-distance map, CTCF ChIP-seq track used to define LE anchor candidates, and PRISM-inferred binding-site types used in the SBS description. **(b)** Comparison between the experimental median-distance map and the LE+SBS prediction; similarity is quantified by the genomic distance-corrected Pearson correlation $r' = 0.70$. **(c-d)** The comparison of the average distance of a specific pair of sites from different models and experiments. (adapted from [17])

potential is used for the repulsive interaction to ensure the polymer self-avoids. Binders interact with the beads via the Lennard-Jones potential, and loop extrusion motors bind to the polymer as harmonic springs. The Langevin thermostat is used to set the temperature. Each simulations run up to 10^8 iterations before reaching equilibrium. An ensemble of 10^3 polymer configurations is generated from parallel simulations to compute the contact matrix, median-distance map, and other properties characterizing the polymer.

The simulation results are then compared with FISH data, in which the median radius of gyration R_g is fitted to the microscopy value to convert the simulation's natural units to the real-world scale. The SBS model (and LE+SBS) gives comparable results in the contact map and median distance map (figure 2.3 b).

2.4. Other chromosome models

Lee et al. [21] introduce a multi-phase-field modeling framework for nuclear architecture, representing the nucleus boundary, chromosome territories, and heterochromatin as continuous fields (with euchromatin defined implicitly as the complement of heterochromatin within each chromosome territory). The different fields interact in a way resembling Landau–Ginzburg theory, with dynamics derived from a Ginzburg–Landau–type free-energy functional (gradient penalties and double-well potentials). They show that inverted nuclear architecture (heterochromatin centralized) and the conventional→inverted reorganization process can be reproduced under physically defined conditions, rather than requiring a specific genotype.

Falk et al. [22] developed an equilibrium block-copolymer bead–spring model of whole chromosomes in which each monomer (~ 40 kb) is assigned to A (euchromatin), B (heterochromatin), or C (pericentromeric constitutive heterochromatin) and interacts via short-range, type-dependent attractions (with optional heterochromatin–lamina attraction), to explain why A/B compartment “plaid” pat-

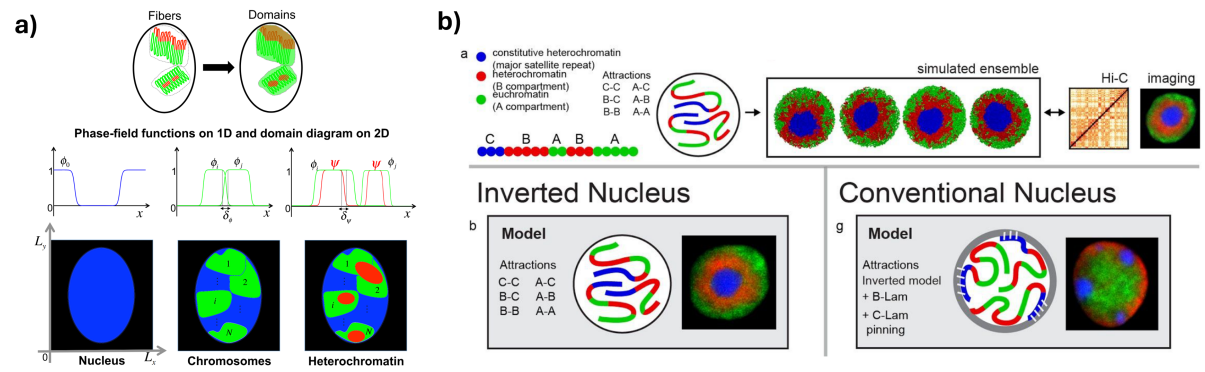


Figure 2.4: (a) The field-based model by Lee et al. [21]. (b) The block-copolymer bead-spring model by Falk et al. [22].

terns persist even when nuclei undergo dramatic radial reorganization (conventional versus inverted architecture).

2.5. Self-consistent field theory

Field-based computer simulation is an alternative strategy to the particle-based MD simulation. Instead of tracking particle positions and velocities, it utilizes the statistical field theory model, which describes the polymer as a continuous density field that varies with position and its conjugate chemical potential fields that encode the pairwise interactions between polymer elements. This method enables efficient computation of equilibrium morphologies at mesoscopic scales. [23–25]

The self-consistent field theory (SCFT) is a powerful theoretical method for studying inhomogeneous polymer systems at equilibrium, firstly introduced by Edwards in the 1960s [26]. One of the most significant applications, proposed by Helfand in 1975 [27], is the prediction of complex ordered nanodomains or microphases formed by block copolymers. This is achieved through a relatively simple description of Gaussian chains with short-range contact interactions, utilizing the Flory-Huggins theory. The system state is obtained by iteratively solving for: first, chain conformations (density) in a chemical potential field, and then updating this field consistently with the polymer density, until self-consistency is reached. The details and formulations of SCFT are introduced in the sections 4.1 and 4.2. [28]

To illustrate the capability of SCFT, we present a SCFT study of the self-assembly behavior of ABC triblock copolymers, serving as a benchmark system, as demonstrated by Wang [29]. The polymer consists of a first 10% contour length block A being hydrophilic with ($\chi_{AS} = 0.5$), followed by a 15% block B and a 75% length block C that are hydrophobic, with χ_{BS} and χ_{CS} ranging from 20 to 60. The χ between blocks ranges from 15 to 35 (see section 4.2 for the detailed explanation of the χ parameter). Depending on these interaction parameters, the copolymers phase-separate from the solution and self-assemble into vesicles. Figure 2.5a shows representative circular vesicles formed by the copolymers, and Figure 2.5b provides a schematic interpretation of the corresponding chain conformations within the mean-field description.



Figure 2.5: (a) The ABC tri-block copolymers self-assemble into vesicles. (b) The scheme of polymer conformation represented in the mean-field polymer model. (adapted from [29])

We reproduced this triblock polymer calculation using our own code as a validation and obtained comparable results. We further investigated this system to analyze its computational convergence and to examine how the SCFT predicts the free energy landscape during phase transitions, as shown in Figures 4.3 and 4.4. We discussed these results in sections 4.4.5 and 4.5.

Methods: Binding sites inferring

Inspired by PRISMR [16] [17] [20] and the open-source MeanFieldChromatin [30], we developed our own inference workflow that converts an experimental Hi-C contact map into a discrete bead "color" sequence along a chromatin polymer. The sequence serves as input to SCFT later in the thesis, where we predict the density and contact pattern.

The core idea behind this algorithm is to iteratively randomly alternate the color of individual beads and evaluate whether the resulting polymer has a contact pattern more consistent with the experimental input. In principle, one can compute the contact map of the new bead sequence via MD simulations for each iteration. However, due to the computational time it costs and the number of iterations needed, it is almost impossible and impractical to perform MD simulations for the Monte Carlo iterations. We employed the mean-field approximation to rapidly create the contact map (covered in Section 3.2). This method is also utilized in the SBS model study by Bianco et al. [17]. It uses statistical mapping from color match/unmatch to empirically predict contact probability.

Note: Some steps and parameters introduced in this chapter may not be the most optimized as they could be. The Hi-C data and the corresponding chromatin properties may vary from case to case. Due to the scope and focus of this project, we didn't test a wide range of data to achieve the "perfect preset".

3.1. Preprocessing

Hi-C contact maps typically span multiple orders of magnitude, with the largest values concentrated near the diagonal, with a power-law decay over genomic distance. To prevent near diagonal elements from dominating the fit, the cost function (section 3.4) is computed at log scale. This transformation yields a more balanced contribution from short- and long-range contacts.

As a first preprocessing step, we therefore transform the Hi-C map to a logarithmic scale and fill the points that contain NaN values with their adjacent values. Depending on the dataset and downstream optimization parameters, it may be beneficial to clip the Hi-C values to a specific range for improved fitting in subsequent steps.

3.2. Mean-field approximation: built contact matrix

To enable Monte Carlo inference over thousands of iterations, we require a fast way to predict the contact matrix based on bead "color" sequence each iteration without running a full MD simulation every time. We therefore use a mean-field approximation which predicts the contact matrix in a statistical way. The contact probability follows the two contact probability profiles measured from MD simulation: coil states and globule states (figure 2.2b). For a pair of candidates, we first calculate their genome distance

$s_{ij} = |i - j|$, then assign the contact probability

$$P_{ij} = \begin{cases} P_{glob}(s_{ij}), & c_i = c_j, \\ P_{coil}(s_{ij}), & c_i \neq c_j. \end{cases}$$

which is consistent with the SBS model intuition that beads of the same binding class are more likely to be bound and exhibit a compact state.

A subtlety is that the experimental Hi-C value can be anything between the coil and globule states' contact probability (and in some cases, outside the range, but this is not considered in this case), rather than a "binary value" of one or the other. To address this issue, we utilized multiple beads to represent a single pixel on the Hi-C map. We then constructed a contact map at a higher resolution, and downscale it to the Hi-C resolution using a Gaussian filter. This procedure effectively averages the values of multiple pairs of beads to obtain a value that matches the Hi-c value.

3.3. Matching power law profile

Hi-C maps can have different genomic bin sizes (resolutions), and the mean-field construction introduces an additional discretization through refined bead representation. As a result, the contour distance matrix used for calculating the contact matrix must be calibrated so that a pixel in the constructed matrix corresponds to the correct genomic separation in the Hi-C map.

We perform this calibration by matching the distance-dependent decay profile of the constructed matrix to that of the Hi-C data. Specifically, we adjust two scalar parameters: (i) The contour zoom factor, which rescales the contour distance, and (ii) the intensity zoom factor, which rescales the overall contact probabilities. These two profiles were tuned so that the power-like decay profile of the constructed matrix matches the Hi-C data. Once determined, these two factors are used in the subsequent steps and remain unchanged in the Monte Carlo iterations, so that it focuses on optimizing the bead color sequence. (It is optional to have the scale calibrated again after a thousand iterations, to allow the scales to be calibrated with a more accurate sequence.)

3.4. Cost function

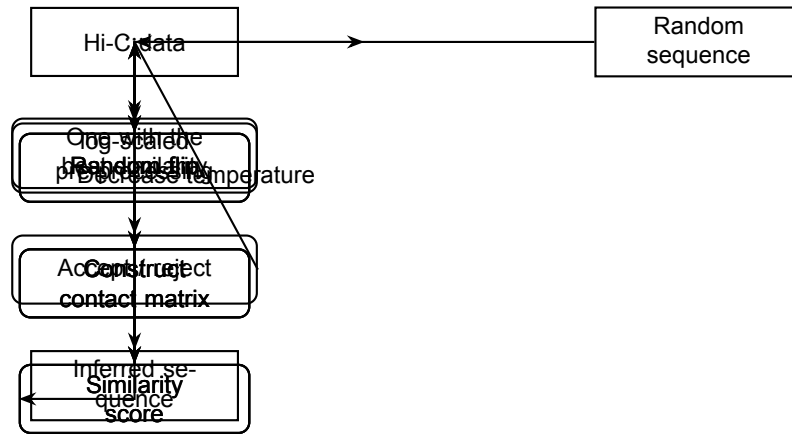
To quantify how well a polymer bead color sequence reproduces the Hi-C pattern, we define a cost function that compares the construct matrix to the preprocessed Hi-C matrix. We use the mean absolute error between these matrices

$$C(A, B) = \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n |A_{ij} - B_{ij}|.$$

A and B represent the constructed contact matrix and the Hi-C matrix, respectively. C is the cost score. Other types of scoring method can be used or combined, for example, Pearson correlation or HiCRep [31].

Additional terms can be added to this cost function to prevent overfitting. One of the simplest options is to add a penalty proportional to the number of distinct colors used in the bead sequence. Other penalties are also possible, depending on the desired prior assumptions, for example, terms that evaluate the clustering behavior of beads with the same color.

3.5. Monte Carlo method and simulated annealing



Starting from an initial random bead sequence, the Monte Carlo process optimizes the sequence over thousands of iterations to minimize the cost function. Each iteration consists of multiple trials. In each trial, a random color is assigned to a randomly selected bead, yielding a candidate sequence and corresponding contact matrix. The cost function is evaluated for each trial, and the candidate with the lowest cost score is chosen as the proposed update for that iteration. This best-of-N trials strategy helps avoid the system from getting stuck in a local minimum.

The score of the best candidate sequence in the current trial, denoted by s_{new} , was compared with the score of the sequence from the previous iteration, s_{old} . The difference $\Delta s = s_{new} - s_{old}$ was then used in a Metropolis acceptance step to decide whether to replace the previous sequence. Specifically, the new sequence was accepted with probability

$$A = \begin{cases} \exp(-\frac{\Delta s}{T}), & \text{if } \Delta s > 0, \\ 1, & \text{otherwise.} \end{cases}$$

where T is the effective temperature.

To implement simulated annealing, the temperature T is initialized at a relatively high value and decreases over iterations. At high T , the algorithm explores the sequence broadly by accepting positive Δs with higher probability; as T decreases, the search refines the best-fit sequence by rejecting most of the uphill moves. This helps the system find the global minima at the beginning, then has a more accurate fitting result at the end. Figure 3.1 summarizes a representative bead color sequence result inferred from the Hi-C map of 28-30 Mb region on chromosome 21 of IMR90 cells [4, 32] through this simulated annealing process, showing the constructed matrix, which exhibits a high similarity to the Hi-C map input.

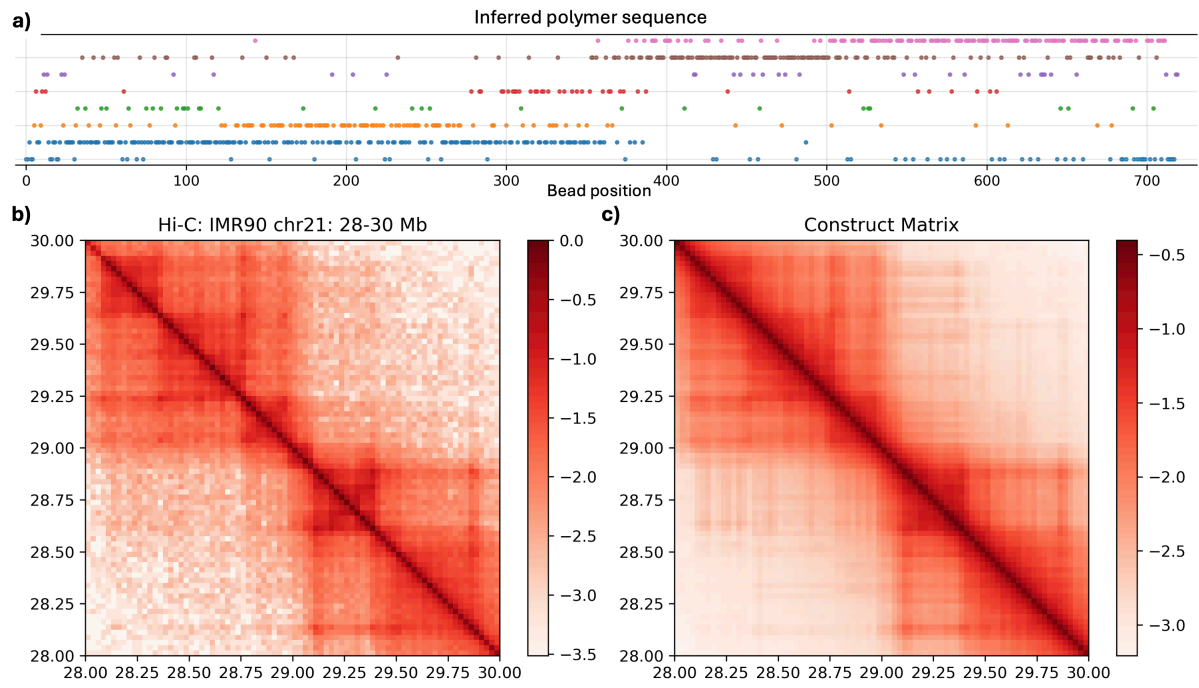
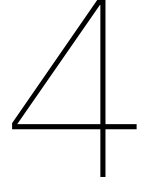


Figure 3.1: **(a)** The bead sequence inferred from **(b)** the Hi-C map shown. **(c)** The contact matrix constructed from the inferred sequence.



Methods: Self-consistent field theory

In this chapter, we introduce the self-consistent field theory (SCFT) framework used throughout this thesis to compute equilibrium chromatin organization. SCFT is a mean-field (saddle-point) approximation in which the polymer is represented by continuous volume-fraction fields that are conjugate to spatially varying chemical-potential fields. The key idea is to iteratively compute polymer conformations in a given field and then update the field based on the resulting polymer density. When the iteration converges, the fields and densities are mutually consistent, and the system is said to be self-consistent.

More concretely, each SCFT iteration consists of (i) solving the modified diffusion equation (MDE) for the chain propagators under the current fields, (ii) computing polymer (and, when included, solvent) volume fractions from the propagators, and (iii) updating the chemical-potential fields using the interaction model (Flory–Huggins) together with the incompressibility constraint. To improve numerical stability, the updated fields are typically mixed with the previous iterate. The converged solution is used to compute observables such as the free energy, contour-resolved density organization, and the simulated contact map for comparison with Hi-C.

4.1. SCFT equations

In SCFT, chain conformations are most commonly modeled using either the Gaussian chain (GC) or worm-like chain (WLC) models. The WLC description retains local orientational information and is therefore useful for stiff or liquid-crystalline polymers, but it requires solving a higher-dimensional propagator (typically in both position and orientation space) and is substantially more computationally expensive; practical implementations often rely on path-integral / orientational-diffusion solvers [33]. In contrast, the Gaussian-chain model treats the polymer as a random walk in which successive steps are uncorrelated. This approximation is appropriate for flexible chains and leads to a diffusion-like description of chain statistics in an external chemical-potential landscape.

In the Gaussian-chain SCFT formalism, polymer statistics in a chemical potential field are expressed in terms of the forward and backward propagators, $q(\mathbf{r}, s)$ and $q^\dagger(\mathbf{r}, s)$, which satisfy the modified diffusion equations (MDEs)

$$\frac{\partial q(\mathbf{r}, s)}{\partial s} = D \nabla^2 q(\mathbf{r}, s) - \omega(\mathbf{r}, s) q(\mathbf{r}, s), \quad (4.1)$$

$$\frac{\partial q^\dagger(\mathbf{r}, s)}{\partial s} = -D \nabla^2 q^\dagger(\mathbf{r}, s) + \omega(\mathbf{r}, s) q^\dagger(\mathbf{r}, s), \quad (4.2)$$

Here $s \in [0, 1]$ is the contour coordinate and $D = b^2/6$, where b is the Kuhn length.

The modified diffusion equation (MDE) is typically solved with free-end boundary conditions along the contour coordinate, in which the initial condition of the propagators are set to $q(\mathbf{r}, 0) = 1$ and $q^\dagger(\mathbf{r}, 1) = 1$. Alternatively, tethered boundary condition can be imposed by fixing one or both sides to prescribed spatial locations.

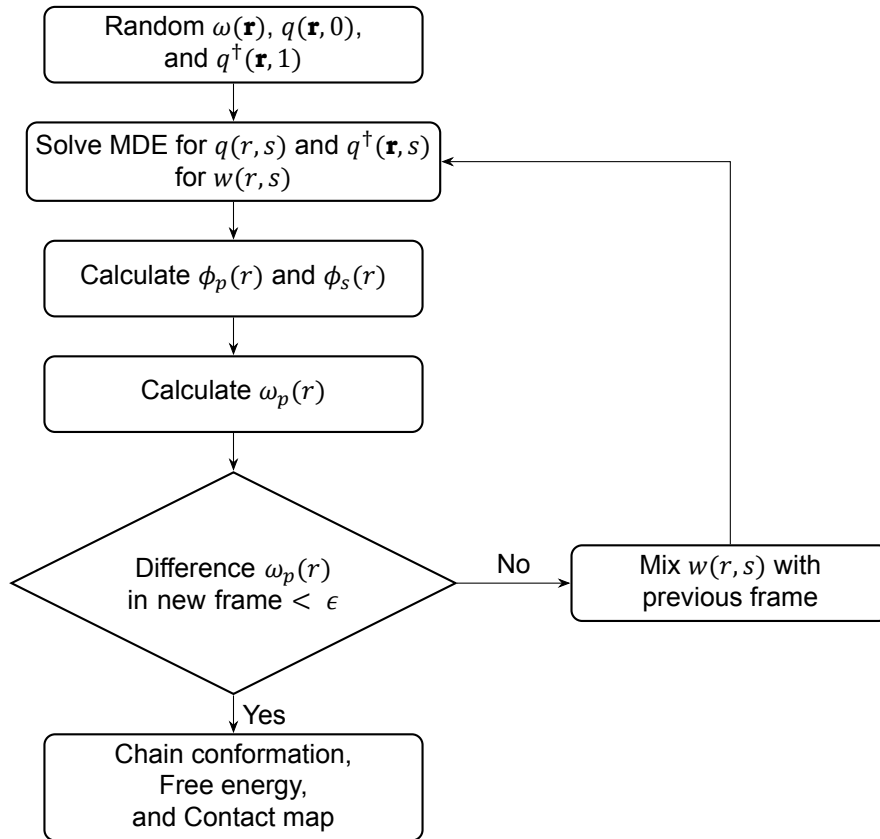


Figure 4.1: Schematic of the SCFT fixed-point iteration used in this thesis. Starting from an initial guess for the chemical-potential fields (and, if needed, propagators), the MDE is solved to obtain forward and backward propagators. Polymer (and solvent) volume fractions are computed from these propagators, the fields are updated using the interaction model and incompressibility constraint, and field mixing is applied to stabilize convergence. The iteration terminates when the field update is smaller than ϵ , after which structural observables (density organization, free energy, and contact map) are evaluated.

The single-chain partition function in the external field is

$$Q[\omega] = \frac{1}{V} \int d\mathbf{r} q(\mathbf{r}, 1) = \frac{1}{V} \int d\mathbf{r} q^\dagger(\mathbf{r}, 0), \quad (4.3)$$

where V is the system volume. The (dimensionless) contour-resolved segment distribution is proportional to the product of the two propagators,

$$\phi(\mathbf{r}, s) = \frac{1}{Q} q(\mathbf{r}, s) q^\dagger(\mathbf{r}, s), \quad (4.4)$$

and the polymer volume-fraction field is obtained by integrating along the contour,

$$\phi_p(\mathbf{r}) = \frac{\bar{\phi}_p}{Q} \int_0^1 ds q(\mathbf{r}, s) q^\dagger(\mathbf{r}, s). \quad (4.5)$$

Here $\bar{\phi}_p$ is the prescribed mean polymer fraction, which ensures $\langle \phi_p \rangle = \bar{\phi}_p$. The volume fraction of a specific block type A is obtained by restricting the contour integral to the corresponding contour segments,

$$\phi_A(\mathbf{r}) = \frac{\bar{\phi}_p}{Q} \int_0^1 ds q(\mathbf{r}, s) q^\dagger(\mathbf{r}, s) \delta_{\text{color}(s)=A} \quad (4.6)$$

where the block-type function $\text{color}(s)$ (the "color" along the contour) is inferred from the Hi-C contact map as described in Chapter 3.

Some implementations avoid explicitly storing a backward propagator by computing a second forward propagator on the reversed contour (or by reversing the field sequence along s). For a homopolymer with $\omega(\mathbf{r}, s) = \omega(\mathbf{r})$, this reduces to the familiar identity $q^\dagger(\mathbf{r}, s) = q(\mathbf{r}, 1 - s)$, giving

$$\phi_p(\mathbf{r}) = \frac{\bar{\phi}_p}{Q} \int_0^1 ds q(\mathbf{r}, s) q(\mathbf{r}, 1 - s), \quad (4.7)$$

But in general, in multi-block settings, one needs to account explicitly for the s -dependence of $\omega(\mathbf{r}, s)$.

The chemical-potential fields are obtained from the interaction model together with incompressibility. For a binary A/B description with a Flory–Huggins interaction,

$$\omega_A(\mathbf{r}) = \chi_{AB} \phi_B(\mathbf{r}) + \eta(\mathbf{r}), \quad (4.8)$$

$$\omega_B(\mathbf{r}) = \chi_{AB} \phi_A(\mathbf{r}) + \eta(\mathbf{r}), \quad (4.9)$$

where N is the degree of polymerization and $\eta(\mathbf{r})$ is a pressure-like Lagrange multiplier field that enforces local incompressibility, we discuss it in the next section. The field $\eta(\mathbf{r})$ is not unique up to an additive constant (gauge-invariant); only its spatial variations are physically relevant.

In the original Flory lattice picture, the Flory–Huggins parameter quantifies the energetic penalty for mixing unlike segments and can be expressed in terms of nearest-neighbor contact energies as

$$\chi_{AB} \equiv \frac{z}{2} \frac{1}{k_B T} (2u_{AB} - u_{AA} - u_{BB}), \quad (4.10)$$

where $u_{\alpha\beta}$ is the interaction energy between two adjacent lattice sites occupied by segments α and β , and z is the lattice coordination number.

4.2. Multi-block polymer in explicit solvent

The nucleus is a crowded environment. DNA together with bound proteins occupies a substantial volume fraction of the nucleoplasm. The remaining space contains water, ions, and a large variety of soluble macromolecules. In this thesis, we coarse-grain these components into an effective, homogeneous "solvent" species and treat the system as an incompressible mixture of solvent and a single multi-block chromatin polymer.

We impose the local incompressibility condition

$$\phi_s(\mathbf{r}) + \sum_p \phi_p(\mathbf{r}) = 1 \quad (4.11)$$

where $\phi_s(\mathbf{r})$ and $\phi_p(\mathbf{r})$ denotes the solvent and polymer block volume fraction respectively.

For an incompressible mixture with Flory-Huggins interactions, the chemical potential field of each polymer type take the form

$$\omega_A(\mathbf{r}) = \sum_{A \neq B} \chi_{AB} \phi_B(\mathbf{r}) + \chi_{AS} \phi_s + \eta(\mathbf{r}) \quad (4.12)$$

where the potential field for block P is based on the Flory-Huggins interaction with other blocks and the solvent. The chemical potential of the solvent follows a similar format:

$$\omega_s(\mathbf{r}) = \sum_B \chi_{BS} \phi_B(\mathbf{r}) + \eta(\mathbf{r}) \quad (4.13)$$

Here, χ_{AB} quantifies the effective incompatibility between two types of polymer blocks, and χ_{AS} is the incompatibility between one type of polymer block and the solvent. Note that the $\omega(\mathbf{r})$ for different blocks on the polymer may differ. Therefore, the field of the polymer is also a function of the contour $\omega_p(\mathbf{r}, s)$.

The solvent concentration field follows the Boltzmann distribution:

$$\phi_s = \frac{\bar{\phi}_s}{Q_s} e^{-\omega_s(\mathbf{r})} \quad (4.14)$$

with the corresponding single solvent particle partition function

$$Q_s = \frac{1}{V} \int d\mathbf{r} e^{-\omega_s(\mathbf{r})} \quad (4.15)$$

Combining equation 4.14 and 4.11 gives $\phi_s(\mathbf{r}) = 1 - \sum_p \phi_p(\mathbf{r})$, and the chemical potential of the solution can be expressed by the concentration of polymer blocks

$$\omega_s(\mathbf{r}) = -\ln(1 - \sum_p \phi_p) + \text{const.} \quad (4.16)$$

which then can be substituted into equation 4.12 subtract equation 4.13 to give the expression for η , a Lagrange multiplier that encodes the incompressibility into the expression for the chemical potential field of the polymer blocks.

$$\eta(\mathbf{r}) = -\ln(1 - \sum_p \phi_p) - \sum_{A \neq B} \chi_{AB} \phi_B(\mathbf{r}) \quad (4.17)$$

Finally, once a self-consistent solution is obtained, the mean-field free energy can be obtained. It consists two parts: an interaction (enthalpic) contribution and an entropic contribution associated with the single chain and solvent partition functions:

$$\begin{aligned} F = \rho_0 \int d\mathbf{r} & \left[\underbrace{\sum_{A \neq B} \chi_{AB}(\mathbf{r}) \phi_A(\mathbf{r}) \phi_B(\mathbf{r}) + \sum_A \chi_{AS} \phi_A(\mathbf{r}) \phi_s(\mathbf{r})}_{\text{interaction part}} \right] \\ & - \underbrace{\rho_0 \int d\mathbf{r} \left[\sum_A \omega_A(\mathbf{r}) \phi_A(\mathbf{r}) + \omega_s(\mathbf{r}) \phi_s(\mathbf{r}) \right] - n_c \ln Q_c - n_s \ln Q_s}_{\text{entropic contribution}} \end{aligned} \quad (4.18)$$

where n_c is the number of copolymer chains in the system, n_s the number of solvent particles, and ρ_0^{-1} is the volume occupied by a monomer or particle. The first part of the expression corresponds to the interaction term of the free energy. The second part is the entropic and field constraint part. The free energy is invariant to a uniform chemical potential shift $\omega^*(\mathbf{r}) \rightarrow \omega(\mathbf{r}) + \omega_0$ [23]. We take ω_0 that satisfies $\sum_i \int d\mathbf{r} w_i(\mathbf{r}) = 0$. This helps prevent the propagator $q(\mathbf{r}, s)$ from exceeding the floating-point limit in numerical computation.

4.3. MDE solver

The equation 4.1 has the structure of a diffusion–reaction equation. $\omega(\mathbf{r})$ acts as the heat source/sink. Two broad numerical approaches are commonly used in SCFT: finite-difference schemes (e.g., Crank–Nicolson) and spectral / pseudo-spectral schemes based on Fourier transforms. Finite-difference methods can accommodate complex geometries and boundary conditions, but for the periodic (or cosine) grids used in this thesis, pseudo-spectral methods are typically more efficient because they evaluate the Laplacian exactly in reciprocal space and leverage highly optimized FFT libraries.

Taking Fourier transform of equation 4.1 with respect to $\mathbf{r}$ yields an evolution equation for the Fourier mode $\hat{q}(\mathbf{k}, s)$:

$$\frac{\partial \hat{q}(\mathbf{k}, s)}{\partial s} = -D(2\pi)^2 |\mathbf{k}|^2 \hat{q}(\mathbf{k}, s) - \omega \widehat{q}(\mathbf{k}, s) \quad (4.19)$$

where $|\mathbf{k}|^2 = k_x^2 + k_y^2 + k_z^2$ and it denotes the discrete spatial frequencies. The term $\omega \widehat{q}(\mathbf{k}, s) = \mathcal{F}\{\omega(\mathbf{r}) q(\mathbf{r})\}$ is first evaluated in real space, followed by the transform.

As an initial approach, we implemented a semi-implicit scheme in s .

$$\frac{q(\mathbf{r}, s + \Delta s) - q(\mathbf{r}, s)}{\Delta s} = D \nabla^2 q(\mathbf{r}, s + \Delta s) - \omega_p(\mathbf{r}, s) q(\mathbf{r}, s) \quad (4.20)$$

which in the Fourier space is

$$\hat{q}(\mathbf{k}, s + \Delta s) - \hat{q}(\mathbf{k}, s) = -\Delta s \left[D(2\pi)^2 (k_x^2 + k_y^2 + k_z^2) \hat{q}(\mathbf{k}, s) + \omega \widehat{q}(\mathbf{k}, s) \right] \quad (4.21)$$

And the $q(\mathbf{r}, s + \Delta s)$ can be written as:

$$\hat{q}(\mathbf{k}, s + \Delta s) = \frac{\hat{q}(\mathbf{k}, s) - \Delta s [\omega \widehat{q}(\mathbf{k}, s)]}{1 + \Delta s D(2\pi)^2 (k_x^2 + k_y^2 + k_z^2)} \quad (4.22)$$

The semi-implicit method has good accuracy for sufficiently small Δs . However, it is not unconditionally convergent due to the field term is explicit; when Δs is too large or $\omega(\mathbf{r})$ is strong, the update may exhibit numerical divergence.

The pseudo-spectrum method, or so-called split-operator method, introduced by Strang [34], is an unconditionally converging method for the equation 4.1. At each forward step, the propagator is transformed by a linear operator $\mathcal{L}$

$$q(\mathbf{r}, s + \Delta s) = e^{\mathcal{L} \Delta s} q(\mathbf{r}, s) \quad (4.23)$$

where the operator $\mathcal{L}$ is composed of two parts: the diffusion operator $\mathcal{L}^D$ and potential field operator $\mathcal{L}^W$:

$$\begin{aligned} \mathcal{L} &= D \nabla^2 - \omega(\mathbf{r}) \\ &= \mathcal{L}^D + \mathcal{L}^W \end{aligned} \quad (4.24)$$

This decomposition closely mirrors imaginary-time propagation in quantum mechanics: diffusion plays the role of the kinetic-energy operator, while $\omega(\mathbf{r})$ serves as a potential-energy landscape.

The second-order expansion in s of the $e^{\mathcal{L} \Delta s}$ equation 4.23 and is equal to

$$\begin{aligned} e^{\mathcal{L} \Delta s} &= 1 + (\mathcal{L}^D + \mathcal{L}^W) \Delta s + \frac{\Delta s^2}{2} [(\mathcal{L}^D)^2 + \mathcal{L}^D \mathcal{L}^W + \mathcal{L}^W \mathcal{L}^D + (\mathcal{L}^W)^2] + O(\Delta s^3) \\ &\approx e^{\mathcal{L}^W \Delta s/2} e^{\mathcal{L}^D \Delta s} e^{\mathcal{L}^W \Delta s/2} \end{aligned} \quad (4.25)$$

Operationally, a single step is implemented as:

1. Multiply in real space: $q(\mathbf{r}) \leftarrow e^{-\omega(\mathbf{r}) \Delta s/2} q(\mathbf{r})$.
2. Transform to Fourier space (FFT/DCT).
3. Multiply each Fourier mode by the diffusion factor: $\hat{q}(\mathbf{k}) \leftarrow e^{-D(2\pi)^2 |\mathbf{k}|^2 \Delta s} \hat{q}(\mathbf{k})$.
4. Transform back to real space.
5. Multiply again in real space: $q(\mathbf{r}) \leftarrow e^{-\omega(\mathbf{r}) \Delta s/2} q(\mathbf{r})$.

4.4. Implementation details and numerical considerations

4.4.1. Contact map

The Hi-C map measures the frequency of pairs (or triplets) of genome loci that physically interact, for example, cross-linked by protein. Super-resolution imaging further indicates that Hi-C contact frequency correlates with physical proximity metrics, such as median spatial distance or the degree of spatial overlap between labeled loci. [5].

In particle-based MD simulations, a contact map is typically calculated by sampling an ensemble of chain conformations and counting pairs of loci whose spatial distance is below a chosen cutoff. However, in SCFT, the polymer is represented by a continuous, ensemble-averaged field rather than an explicit chain location. As a consequence, the definition of "contact" is unclear.

To define an SCFT analogue of a contact map, we use the spatial overlap of the probability of the contour-resolved polymer segments. Let $P(i, j)$ denote the contact probability between contour indices i and j :

$$P(i, j) = \frac{1}{V} \sum_{x=0}^{nx} \sum_{y=0}^{ny} \phi_p(\mathbf{r}, i) \phi_p(\mathbf{r}, j) \quad (4.26)$$

Intuitively, $P(i, j)$ is large when segments i and j tend to localize within the same spatial regions in the SCFT ensemble. We found that this overlap map exhibits a good correlation with the Hi-C map, as discussed in section 5.2.

4.4.2. Boundary conditions

The pseudo-spectrum method naturally encodes periodic boundary conditions. This is not the case for chromatin in the nucleus of the cell. We impose a no-flux boundary condition to make this single-chain-in-the-box scenario physically appropriate.

A straightforward is to set the $\omega_p(\mathbf{r}, s)$ at the boundary to infinity. In practice, this approach can still lead to numerical "leakage" when the contour discretization step is relatively large, because diffusion in the propagator update can transport probability mass across the boundary within a single contour step.

Instead, we implemented a strictly no-flux boundary condition using a discrete cosine transform (DCT) formulation of the pseudo-spectral update. The DCT corresponds to an even extension of the field $\hat{q}(\mathbf{r}, s)$, which enforces a zero normal derivative at the boundary and therefore guarantees zero flux. To preserve GPU acceleration and remain compatible with CuPy's FFT backend, we implement the DCT via an even expansion of the propagator in real space followed by an FFT-based evaluation; although this includes redundant sine components that evaluate to zero, it provides a practical and efficient implementation on the GPU.

However, this perfect no-flux boundary condition introduced boundary-driven artifacts: the polymer preferentially accumulates near edges and corners, thereby reducing the mixing energy penalty with the solvent. In studies that explicitly consider polymer–boundary interactions, this effect is often controlled by adding a boundary interaction potential (for example, screened electrostatic interactions) to the chemical-potential field. In this project, we do not attempt to model detailed chromatin–nuclear-envelope interactions, given the coarse-grained resolution and the system scale (2 Mbp). We applied a trivial boundary condition where the boundary potential is set to $\chi_{PS}N$, the same value as if the boundary is pure solvent $\phi_s(\text{boundary}) = 1$. This effectively sets the boundary so as not to affect polymer conformation.

4.4.3. Parameter normalization

It is conventional to normalize the contour length to 1 and scale other parameters accordingly. Here, we demonstrate how to normalize a chain with N_0 segments (contour length), each of length $b = 1$. We introduce the parameter N that has the same concept and value as N_{seg} , just for the convenience of the math expression. It's important to note that the interaction strength χN_0 and the morphology remain

unchanged through this re-scaling.

$$\begin{aligned}
 N_{seg} = N_0 & \rightarrow N_{seg} = 1 \\
 N = 1 & \rightarrow N = N_0 \\
 b = 1 & \rightarrow b = \sqrt{N} \\
 L = L_0 & \rightarrow L = L_0 \\
 \text{eq. 4.12} & \rightarrow \omega_A(\mathbf{r}) = \sum_{A \neq B} \chi_{AB} N \phi_B(\mathbf{r}) + \sum_B \chi_{BS} N \phi_S + \eta(\mathbf{r}) \\
 \text{eq. 4.14} & \rightarrow \phi_s = \frac{\bar{\phi}_s}{Q_s} e^{-\frac{\omega_s(\mathbf{r})}{N}}
 \end{aligned} \tag{4.27}$$

The Kuhn length b and box dimension L can be further rescaled by:

$$\begin{aligned}
 b = \sqrt{N} & \rightarrow b = 1 \\
 L = L_0 & \rightarrow L = \frac{L_0}{\sqrt{N}}
 \end{aligned} \tag{4.28}$$

4.4.4. GPU acceleration

To keep the implementation accessible and easy to integrate with downstream analysis and visualization, the SCFT codebase in this project was written in Python. The computationally intensive components—solving the modified diffusion equation (MDE), evaluating densities and chemical-potential fields, and computing the free energy—were implemented in a fully vectorized manner. This avoids explicit Python loops over the spatial grid, which would otherwise dominate runtime.

NumPy is a highly optimized package for performing array operations on the CPU, but when it comes to large arrays (e.g., 256×256), performance is limited by memory bandwidth, the inability to fully utilize CPU cores, and the CPU's own speed. To further accelerate computation, we use CuPy, which enables GPU array operations via the CUDA Toolkit. It is almost fully compatible with Numpy and remains easily accessible, making it suitable for running on computers without a GPU. On a PC equipped with an i7-13700K and an RTX 4080, the GPU implementation accelerated a 2D simulation of a 256×256 grid by a factor of 11 (for $ns = 100$, the code runs at 23 iterations per second) compared to Numpy. The speedup is even more pronounced in 3D simulation: for a 128^3 grid, the GPU is 73 times faster than the CPU.

Furthermore, for parameter sweeps reported in section 5.1, we additionally parallelized simulations across multiple runs. When using a grid size of $nx = ny = 128$, a single simulation does not fully utilize the GPU. The joblib package was used to run three simulations concurrently, further accelerating the parameter sweep.

4.4.5. Field mixing and convergence behavior

Because SCFT is solved as a fixed-point iteration, directly replacing the chemical-potential field with the newly computed update can lead to oscillations or numerical divergence, especially when the system is close to an instability or when strong segregation starts to develop. To stabilize the iteration, we apply field mixing (also called under-relaxation), in which the updated polymer field is combined with the field from the previous iteration:

The simple mixing method is used to update $\omega_p(\mathbf{r}, s)$ each iteration

$$\omega_p^{i+1}(\mathbf{r}, s) = \theta \omega_p^{\text{updated}}(\mathbf{r}, s) + (1 - \theta) \omega_p^i(\mathbf{r}, s) \tag{4.29}$$

where θ is the mixing parameter of how aggressively the field is updated each iteration. In this study, θ is usually set to $\theta = 0.01$ for the chromatin, and $\theta = 0.05$ for the benchmark ABC tri-block polymer. Too large θ causes instability, and the result may not converge.

Figure 4.2 illustrates a representative convergence trajectory of the ABC triblock polymer benchmark. Starting from an initial condition in which $\omega_p(\mathbf{r}, s)$ is a small random field with the order of 10^{-3} .

The free energy often changes slowly during the early iterations. After several dozen iterations, the free energy decreases rapidly and then approaches a stable value, indicating convergence to a self-consistent solution. The corresponding evolution of the polymer density field is shown in Figure 4.3. The vesicle-like morphology becomes apparent primarily during the iterations in which the free energy drops steeply.

It's important to stress that this morphology "development" only shows how the equilibrium state is calculated computationally. It does not indicate the time evolution of the system, since the SCFT method only describes a system at equilibrium.

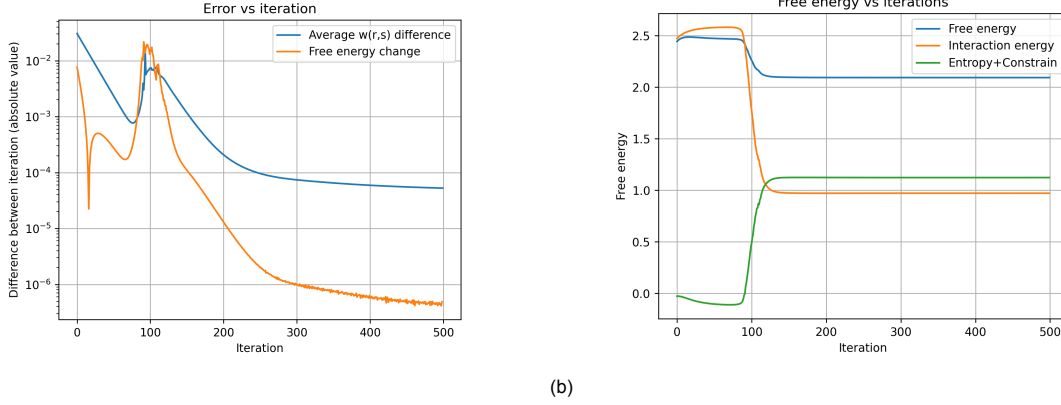


Figure 4.2: **(a)** The free energy change and the average of difference between $\omega_p(\mathbf{r}, s)$ matrix each iteration. For the ABC tri-block copolymer simulation shown in Figure 2.5a. **(b)** The entropic and enthalpic part of the free energy in equation 4.18. The free energy decreases as the polymer's morphology develops.

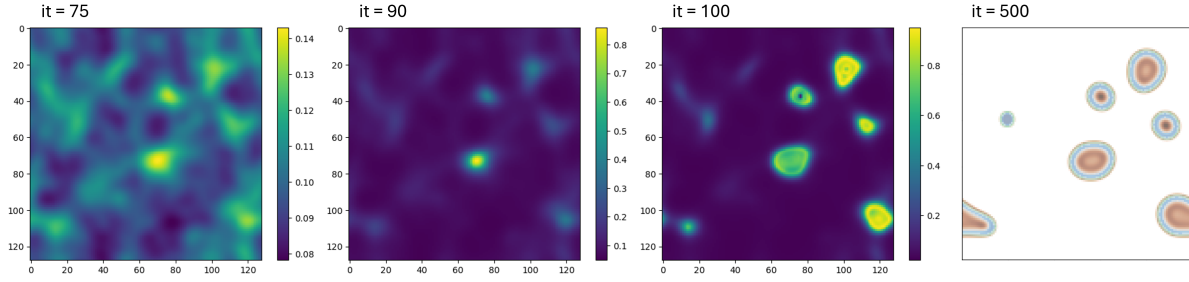


Figure 4.3: The polymer density changes with the iterations. During the 75th to 100th iterations, the morphology of self-assembled polymer vesicles developed. After 500 iterations, it produces a result comparable to Figure 2.5a. $N = 20$, $\phi = 0.1$, $\chi_{AS}N = 0.5$, $\chi_{BS}N = \chi_{CS}N = 10$, $\chi_{AB}N = \chi_{AC}N = \chi_{BC}N = 15$, $L = 100$.

We also attempted to utilize the Anderson mixing method [35] to accelerate convergence. In typical SCFT applications, Anderson mixing is applied after an initial phase of simple mixing, once the iteration has entered a smoother convergence regime. However, for the chromatin polymer with a complex block sequence, the most time-consuming part of the computation is the early "morphology formation" stage under simple mixing. In our case, applying Anderson mixing later did not reduce this dominant cost, and we therefore did not use it in the production simulations.

4.5. Free energy and phase transition

A key advantage of SCFT is that it provides the free energy of the polymer system, as given by equation 4.18. In addition to predicting equilibrium density organization, this enables a quantitative analysis of phase behavior and transitions by tracking how the free energy and its contributions change as interaction parameters are varied.

Here, we demonstrate this idea using the ABC triblock co-polymer that we introduced earlier. We keep the same parameter set as in the example of Figure 4.3, except that we fix $\chi_{BS} = 10$, and vary

the χ_{CS} from 18 to 35. Physically, increasing $\chi_{CS}N$ strengthens the effective repulsion between block C and the solvent, driving self-assembly.

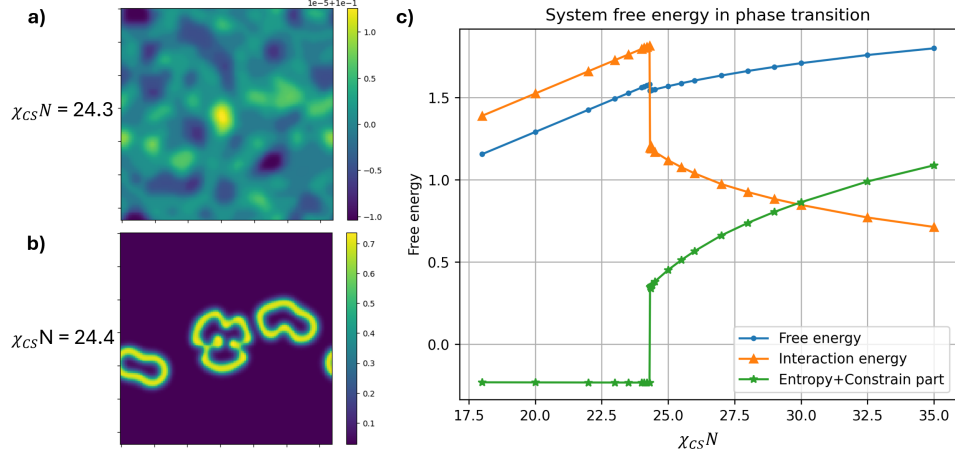


Figure 4.4: **(a)** Polymer density is uniform (with fluctuation at the order of 10^{-5}) at $\chi_{CS}N = 24.3$. **(b)** Polymer phase separate with solvent and form structured vesicles at $\chi_{CS}N = 24.4$. **(c)** Enthalpic, entropic, and total free energy of the system. The first order phase transition happens at C, $\chi_{CS}N = 24.4$

As shown in Figure 4.4(a,b), the system undergoes a sharp phase transition as the hydrophobicity of segment C increases from $\chi_{CS}N = 24.3$ to 24.4. At $\chi_{CS}N = 24.3$, the polymer solution remains essentially homogeneous. Increasing $\chi_{CS}N$ by only 0.1 to 24.4 drives phase separation and self-assembly into an ordered vesicle-like structure.

Figure 4.4c shows the thermodynamic signature of this transition by decomposing the free energy into enthalpic (interaction) and entropic contributions. In the homogeneous regime, the enthalpic contribution increases approximately linearly with $\chi_{CS}N$, whereas the entropic contribution is nearly constant. Once $\chi_{CS}N$ crosses the critical value, the entropic contribution exhibits an abrupt increase, reflecting the entropy penalty associated with forming an ordered, segregated morphology. In contrast, the enthalpic contribution decreases across the transition because segment C becomes sequestered in the vesicle interior, thereby reducing unfavorable C–solvent contacts.

At phase transition, the entropy "jump" and enthalpy "drop" are approximately equal in magnitude, so the total free energy remains nearly continuous across the transition; the small residual drop is attributed to numerical error. For $\chi_{CS}N$ above the critical point, the entropic contribution continues to increase with $\chi_{CS}N$, while the enthalpic contribution decreases. The total free energy satisfies $\partial F / \partial (\chi_{CS}N) > 0$ throughout, but its slope changes abruptly at the critical point. This discontinuity in the first derivative is consistent with a first-order phase transition in the SCFT description.

5

Results

With the SCFT method established and the chromatin binding site sequence inferred from Hi-C, we simulated the conformation of a 2 Mbp chromatin segment and obtained its corresponding contact map (Figure 5.1a). It has a pattern that qualitatively reproduces key patterns in the Hi-C data (Figure 5.1b). In addition to contact patterns, SCFT yields a density field of the polymer for each block type. Figure 5.1c shows the predicted density distribution, where the colored clouds represent the localization of different chromatin blocks. The contact map also exhibits the expected genomic distance-dependent decay of the contact probability. Figure 5.1d compares the decay profile of the normalized SCFT overlap map with that of the Hi-C map.

In this chapter, we quantify how the SCFT model predictions depend on input parameters by performing parameter sweeps. The goal is to find parameters that give results consistent with biological systems, and more importantly, elucidate the physics mechanisms behind these results and chromatin organization.

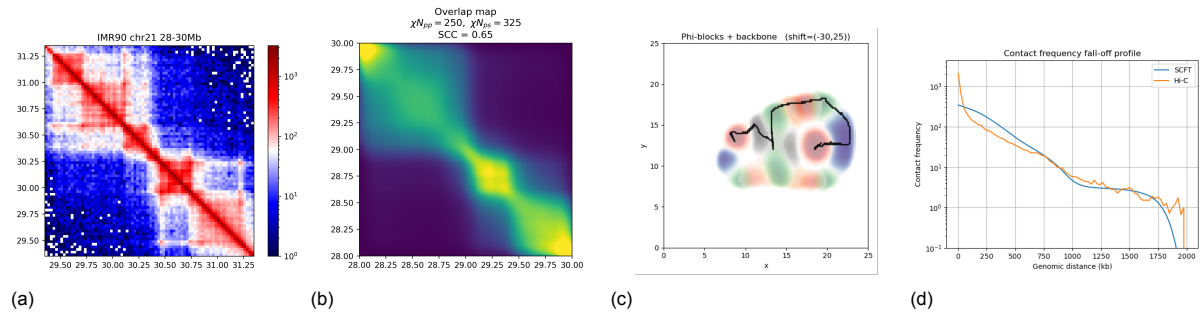


Figure 5.1: (a) Ensemble Hi-C contact frequency matrix for the 2-Mb genomic region binned at 25-kb resolution, data from [32]. (b) The overlap map (contact matrix in SCFT) of the polymer conformation computed in SCFT. (c) The spatial distribution of the polymer. The color clouds correspond to the concentration of different polymer blocks. The black line represents the most likely chain conformation. (d) The comparison of the decay profile between the overlap map result and the Hi-C contact map. The overlap map is normalized to the same level as the Hi-C map by its mean value.

5.1. Parameter sweep and free energy

In SCFT and related Flory mean-field descriptions, the χN parameter characterizes the polymer phase and conformation. First, we focus on two parameters: $\chi_{ps}N$, representing polymer-solvent incompatibility, is an equivalent to "binding strength" between polymer and binder that excludes solvent in the SBS model. $\chi_{pp}N$, representing incompatibility between different polymer blocks, which drives compartmentalization.

We performed a parameter sweep over the combinations of these two parameters: with $\chi_{ps}N$ ranging from 200 to 450 and $\chi_{pp}N$ ranging from 200 to 550 To illustrate the resulting behaviors, Figure 5.2

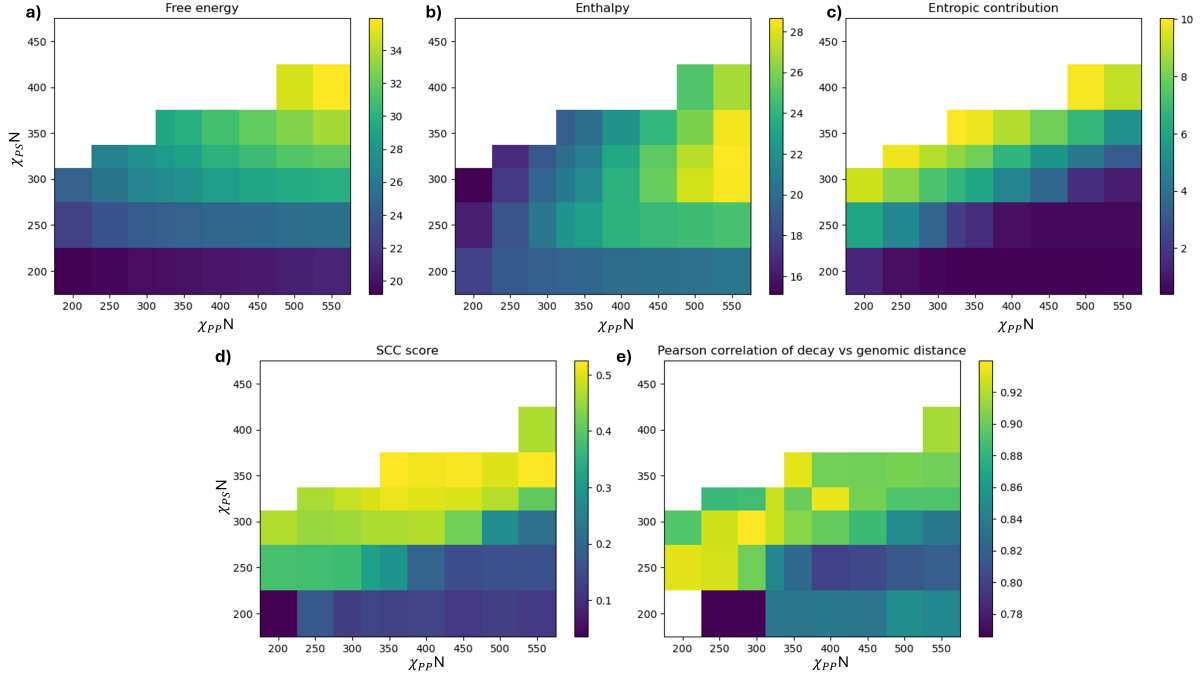


Figure 5.2: Results from the parameter sweep of $\chi_{ps}N$ and $\chi_{pp}N$. Energy landscape of (a) free energy, (b) interaction energy, and (c) energy from entropy. (d) SCC score. (e) Pearson correlation of the log of the power-like decay with genomic distance.

presents the free energy landscape (total, enthalpic, and entropic contributions) as heatmaps, together with the contact map similarity score and the Pearson correlation of the log-decay profile. For certain parameter combinations, especially when $\chi_{ps}N$ is too large, the results failed to converge and were therefore left blank in the heatmaps.

The polymer density maps (Supplement Figure A.1) show two trends: as $\chi_{ps}N$ increases, the polymer begins to phase separate from the solvent, and as $\chi_{pp}N$ increases, the polymer compartmentalization develops. Consistent with this observation, the free energy landscape in the figure 5.2 also implies the phase of the polymer. At low $\chi_{ps}N$, the enthalpy increases with $\chi_{ps}N$, till at some point it starts to decrease with $\chi_{ps}N$, and this is the same point where the entropic part of free energy increases significantly. This indicates the "binding sites" are folded into compartments, similar to the phase transition demonstrated in Figure 4.4 in section 4.5.

However, the behavior observed here differs from the ABC triblock benchmark in an important way. In the present chromatin model, all the blocks share the same $\chi_{ps}N$ and $\chi_{pp}N$ values, such that compaction and compartmentalization emerge from uniform polymer-solvent and polymer-polymer incompatibilities. By contrast, in the latter case, different $\chi_{ps}N$ values across blocks drive the polymer to self-assemble into a highly ordered structure, with the hydrophobic block C fully enclosed within a vesicle. This difference is also reflected in the free energy landscape: As shown in Supplement Figure A.3, increasing $\chi_{ps}N$ in the chromatin model leads to a smooth crossover: $\partial H/\partial(\chi_{ps}N)$ and $\partial S/\partial(\chi_{ps}N)$ change their sign smoothly. On the contrary, the ABC triblock polymer exhibits a critical point of phase transition, where the enthalpy and entropy change abruptly, and the total free energy shows a kink (i.e., a discontinuity in its derivative), consistent with a first-order phase transition.

5.2. Contact map comparison

Figure 5.3 shows the contact maps obtained across $\chi_{ps}N$ and $\chi_{pp}N$. At large $\chi_{ps}N$ but very small $\chi_{pp}N$, the polymer phase separated from the solvent, but the contact map remains largely featureless, indicating that global compaction alone is insufficient to produce Hi-C like structure. It's only when both the $\chi_{ps}N$ and $\chi_{pp}N$ are high enough that the contact map begins to exhibit patterns that qualitatively resemble the Hi-C map.

To quantify the similarity of the structures between the overlap map in SCFT and the Hi-C map, we used HicRep [31, 36] to calculate the stratum-adjusted correlation coefficient (SCC). We also calculated the Pearson correlation between the power-like decay with genomic distance. The decay profile plots are included in the Supplementary Figure A.2.

The SCC scores and the Pearson correlation of the parameter sweep are shown in figures 5.2 d and e. Both metrics identify a consistent high-scoring region in the $\chi_{ps}N$ - $\chi_{pp}N$ plane, indicating that the parameter regimes that best reproduce 2D contact structure and global distance scaling. The polymer conformation and the resulting contact map emerge from the combined effects of chain connectivity (random-walk statistics), polymer-solvent interactions, and Flory-Huggins incompatibility between different polymer blocks (Though the underlying biology is more naturally framed in terms of self-affinity within the polymer, these effects can be represented here as effective repulsive interactions).

Interestingly, at very large χN , despite the patterns on the contact map looking sharper, the polymer is actually over-squeezed by its incompatibility with the solvent, and therefore shows lower scores.

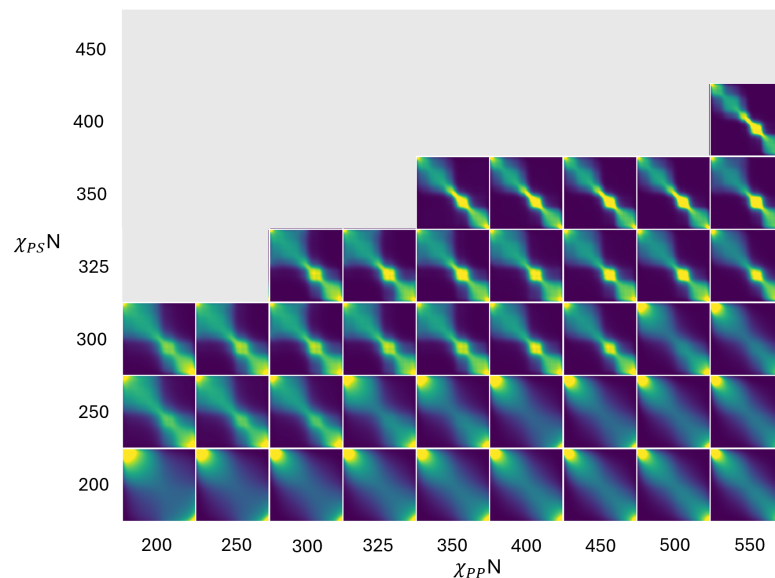


Figure 5.3: The contact maps of the simulated results from the parameter sweep of $\chi_{ps}N$ and $\chi_{pp}N$.

5.3. Density distribution to chain conformation

In SCFT, the colored blobs in figure 5.1c represent the spatial density distribution of the polymer. This total density $\phi(\mathbf{r})$ is obtained by integrating its contour segment concentration $\phi_p(\mathbf{r}, s)$ over the contour coordinate (equation 4.4 and 4.5).

While $\phi(\mathbf{r})$ is useful to identify compartments and phase boundaries, it lacks information regarding the contour coordinate. To better visualize $\phi_p(\mathbf{r}, s)$, we plot one-dimensional slices along certain directions on the polymer blob. The Figure 5.4 shows two of these "contour-space" maps. For each colored horizontal line in real space (leftmost panel), we plot the segment density as a function of spatial position (horizontal axis) and contour coordinate s (vertical axis).

The conformation of the polymer can be visualized intuitively from this graph, which shows how different parts of the polymer localize in space. For a certain region on the polymer, the spatial distribution is split into 2 or multiple parts. This indicates that the polymer might have multiple configurations at equilibrium states. This split distribution is also discussed in section 6.2.

To visualize the dominant localization mode in the contour-resolved density, we plot the density ridge curve in Figure 5.1c by connecting the positions where each polymer contour segment has its maximum density. This curve should be interpreted as a ridge of an ensemble-averaged SCFT field, not as a unique single-chain conformation.

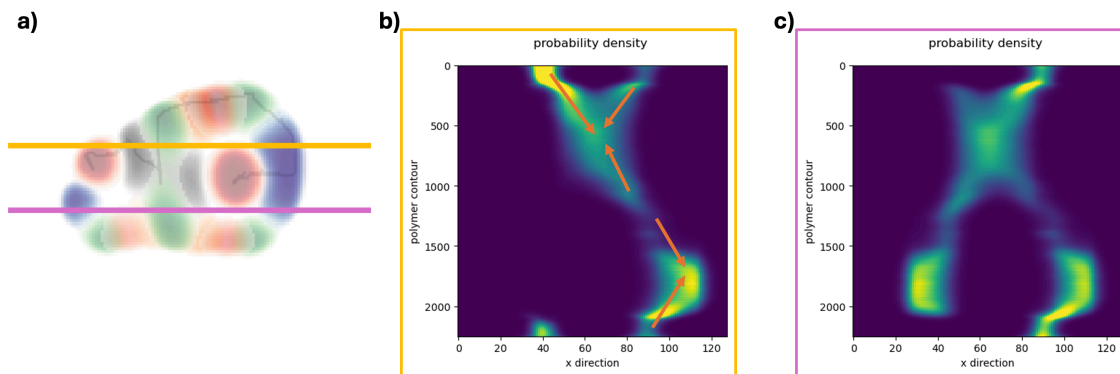


Figure 5.4: One-dimensional slices of the contour probability density. The horizontal axis is the x-position coordinate, and the vertical axis is the polymer contour coordinate. **(a)** The positions of the two slices relative to the polymer. **(b)** The slice corresponding to the upper line in **(a)**. **(c)** The slice corresponding to the lower line in **(a)**.

5.4. Radius of Gyration

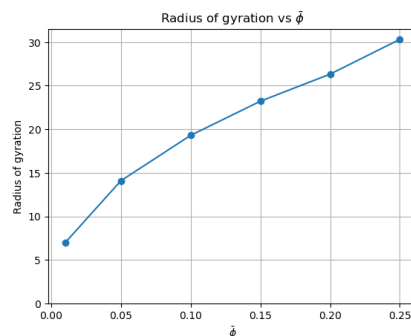


Figure 5.5: The radius of gyration as a function of $\bar{\phi}$

To study how the average polymer density parameter $\bar{\phi}$ affects the conformation, we conducted another parameter sweep, varying $\bar{\phi}$ from 0.01 to 0.25. For each result, we computed the polymer radius of gyration from the total density field $\phi(\mathbf{r})$.

Across this range, the polymer remains phase-separated from the solution (Supplementary Figure A.4), and the radius of gyration increases approximately linearly with $\bar{\phi}$, as shown in Figure 5.5, and the contact map remains relatively unchanged (Supplementary Figure A.5).

5.5. A/B compartmentalization

Thus far, we have used uniform interaction parameters, $\chi_{ps}N$ and $\chi_{pp}N$, for all polymer blocks. Under this assumption, all polymer blocks experience the same effective "solvent repulsion" and the same degree of mutual incompatibility. As a result, internal localization within the phase-separated polymer was not clearly observed.

A straightforward extension is to introduce block-dependent interaction parameters, based on chromosome A/B compartmentalization. Compartments A are generally active, gene-rich, and less compact, whereas compartments B are more inactive and compact. Compartment annotations can be obtained by PCA eigenvector analysis of the normalized Hi-C map [37] and epigenetic tracks measured by Chip-seq, including histone modifications [38, 39].

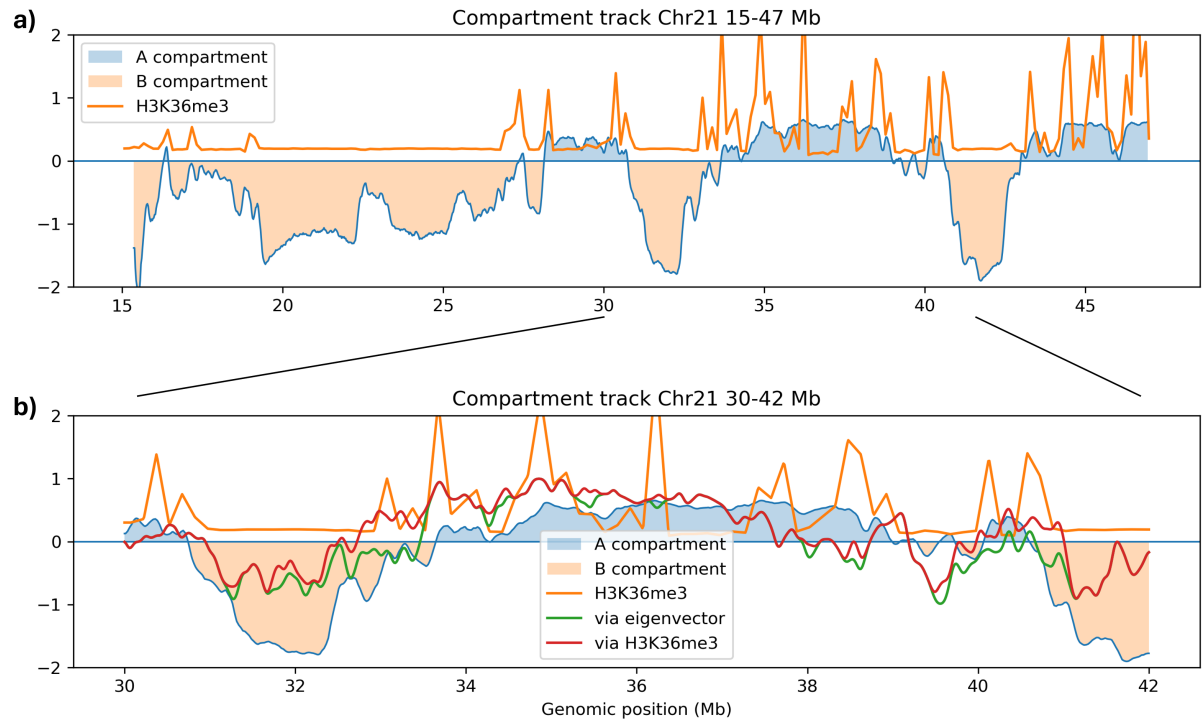


Figure 5.6: **(a)** A/B compartment eigenvector and Gaussian-smoothed (150 kb window) H3K36me3 Chip-seq signal on 15-47 Mb region on Chromosome 21 of IMR90 cell. **(b)** The same analysis is shown for the 30-42 Mb region. Also shown are the compartments curve reconstructed from the inferred polymer-block sequence using either the eigenvector-based annotation (green) or the H3K36me3-based annotation (red).

We performed eigenvector-based compartment analysis on the 15-47 Mb region on Chromosome 21 (Figure 5.7a) in IMR90 cells [4, 32] using cooler [40]. In the resulting eigenvector, as shown in Figure 5.6a, the regions with elevated H3K36me3 signal [41] largely coincide with A compartments, consistent with H3K36me3 being associated with transcriptionally active gene bodies [42]. We then transferred the A/B compartment information to the polymer blocks inferred for the Chr21: 30-42 Mb region (Figure 5.6b), based on the eigenvector-based compartment analysis or the threshold H3K36me3 signal, as follows:

$$C_{\text{PCA}}(s) = \begin{cases} 1, & \text{if } s \text{ is A compartment} \\ -1, & \text{if } s \text{ is B compartment} \end{cases}$$

or

$$C_{\text{H3K36me3}}(s) = \begin{cases} 1, & \text{if } L_{\text{H3K36me3}}(s) > \text{threshold} \\ -1, & \text{if } L_{\text{H3K36me3}}(s) < \text{threshold} \end{cases}$$

$$I_m = \sum_s C(s) \delta(\text{color}(s), m)$$

$$m \text{ is } \begin{cases} \text{A compartment,} & \text{if } I_m > 0 \\ \text{B compartment,} & \text{if } I_m < 0 \end{cases}$$

where $C(s)$ is the index indicating if a genome coordinate s is in compartment A or B. m is the polymer block (binding site) type. $L_{\text{H3K36me3}}(s)$ is the Gaussian-smoothed H3K36me3 Chip-seq signal.

Figure 5.7b shows the inferred polymer blocks annotated by their assigned A/B compartment. As a validation, we reconstructed the compartmentalization curve by mapping each coordinate s to the compartment label of its block type, assigned by eigenvector analysis or Chip-seq. As plotted in Figure 5.6b, both the reconstructed curves reproduce the large-scale compartment structure reasonably well.

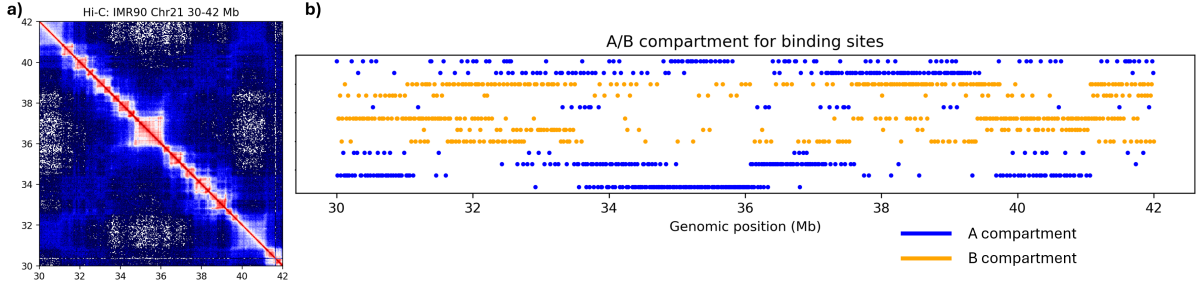


Figure 5.7: **(a)** Hi-C contact map for the 15-47 Mb region on Chromosome 21 in IMR90 cell. **(b)** Polymer blocks inferred from the Hi-C map, with the color annotated by compartment assignment.

To reflect the open or compact state of the chromosome corresponding to the A or B compartments, different $\chi_{ps}N$ and $\chi_{pp}N$ values were assigned to polymer blocks as follows:

$$\begin{aligned}\chi'_{AA} &= \chi_{pp} - \chi_{\text{offset}}, & \chi'_{AB} &= \chi_{pp}, & \chi'_{BB} &= \chi_{pp} + \chi_{\text{offset}}, \\ \chi'_{AS} &= \chi_{ps} - \chi_{\text{offset}}, & \chi'_{BS} &= \chi_{ps} + \chi_{\text{offset}}.\end{aligned}$$

With $\chi_{\text{offset}} > 0$, A-type blocks experience weaker effective repulsive interaction from the solvent and reduced segregation interaction with other blocks, where B-type blocks experience the opposite trend. To investigate if this A/B compartment-dependent χ_{offset} affects the polymer conformation, we performed a parameter sweep over $\chi_{\text{offset}}N$ ranging from -30 to +30.

As shown in Figure 5.8a, the internal organization of the polymer changes as $\chi_{\text{offset}}N$ shift, while the radius of gyration remains relatively constant (Supplementary Figure A.7). Figure 5.8b highlights the localization of blocks annotated as A or B compartments. For negative $\chi_{\text{offset}}N$, B compartments localized toward the periphery of the phase-separated polymer, whereas for positive $\chi_{\text{offset}}N$, they localized towards the center. Conversely, A-compartment blocks tend to enrich in the outer layer, consistent with the expectation that A-compartment chromatin is more open than B-compartment chromatin under $\chi_{\text{offset}}N > 0$.

Unexpectedly, the corresponding contact maps show a trend that does not match this intuitive picture. At negative $\chi_{\text{offset}}N$, the contact map exhibits a pronounced pattern along the diagonal with high intensity, indicative of strong short-range contacts along the polymer contour. As $\chi_{\text{offset}}N$ increases, the pattern along the diagonal progressively fades, while the off-diagonal contacts become more prominent, suggesting an increase in long-range contact.

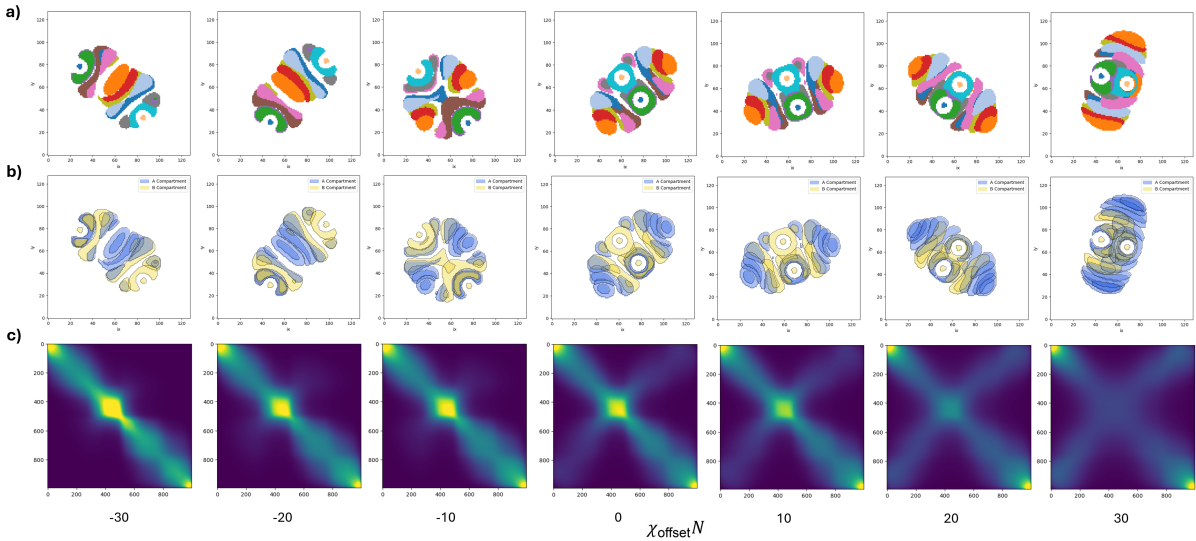


Figure 5.8: **(a)** Polymer density distributions for the parameter sweep, colored by inferred polymer block type. **(b)** The same conformations with blocks grouped by compartment annotation (A vs B). **(c)** Contact maps computed from the corresponding polymer conformations for each $\chi_{\text{offset}}N$ condition. The supplement Figure A.8 shows all the data points in the parameter sweep.

5.6. Simulation in 3D

Leveraging GPU acceleration together with the pseudo-spectral solver, we extended the SCFT computations to three dimensions. Using the same set of parameters as in Figure 5.8, with $\chi_{\text{offset}}N = 10$, the 3D simulation yields a contact map (Figure 5.9a) that is qualitatively consistent with the 2D result.

To visualize the simulated polymer conformation, we render the isosurface of the total polymer block density and the contour-resolved isosurfaces for individual polymer contour segments, as shown in Figure 5.9 b and c. To better visualize the contour-resolved localization, we additionally plotted a curve tracing the centers of the segment isosurfaces, providing a compact representation of the inferred chain path through space.

Due to time constraints, we did not implement a quantitative comparison of the simulated conformations to experimental FISH measurements.

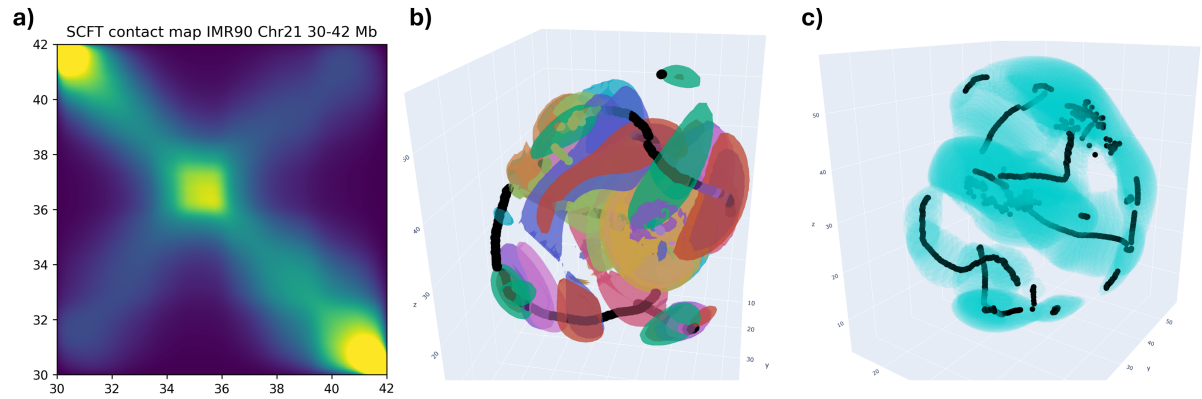


Figure 5.9: **(a)** Contact maps computed from 3D SCFT. **(b)** 3D isosurface of the polymer block density, colored by inferred polymer block type. **(c)** 3D isosurface rendering of contour-resolved segment densities, with an overlaid curve connecting the segment-isosurface centers to indicate the inferred chain path.

Discussion

6.1. Next steps: connect entropy with single-molecule force

A crucial aspect to extend the present result is the interpretation of the Flory-Huggins interaction parameters in biological terms. In SCFT, the Flory χ parameters represent volumetric mixing energy, whereas loop extrusion motors and chromatin-protein interactions are described at the single-molecule level by stepping force and binding energies. Establishing a strict analytical conversion from χ to loop extrusion is therefore nontrivial.

Nevertheless, several studies have shown that the effective Flory χ parameter can be extracted from MD simulations [43–45]. In future work, such a connection could be used to relate the χ used in SCFT, which yields the optimum result, to the single-molecule stepping force measurement of loop extrusion motor [10]. We hope it can be used to explain how energy consumption during chromosome organization reduces the system’s entropy.

6.2. Breakdown of mean-field SCFT in single-chain modeling

Despite the reasonably good contact map result generated by the SCFT method, the obtained spatial density organization does not closely match the structures reported from FISH microscopy [5] and the SBS model [19]. We attribute this discrepancy to two primary limitations of the current implementation.

The first limitation is that our simulations were restricted to a 2 or 12 Mbp-long chromatin segment, thereby neglecting the rest of the chromosome and other chromosomes in the crowded nuclear environment. In vivo, long-range and interchromosomal contacts, as well as chromatin-associated proteins, including histone-mediated packaging and other architectural factors, are present. Omitting these contributions can produce a polymer organization that differs from the actual chromatin conformation.

Second, and more fundamentally, is the limitation of the mean-field SCFT approach in single-chain prediction. In SCFT, the probability density function $\phi_p(\mathbf{r}, s)$ is interpreted as a polymer concentration, based on which the chemical potential field $\omega(\mathbf{r})$ is calculated using the Flory-Huggins interactions. The breakdown lies in this translation of meaning: the probability density is a single-chain property, while the concentration doesn’t limit the number of chains. Multiple distinct chains can contribute to the same mean field, and SCFT can therefore yield multi-modal contour densities, which we observed in Figure 5.4.

This limitation can be illustrated by the ABC triblock copolymer system we discussed in section 2.5, 4.4.5, and 4.5. The formulation of the propagator $q(\mathbf{r}, s)$ and modified diffusion equation (eq. 4.1) is derived based on a Gaussian chain model. However, the vesicle structures shown in the results (figure 4.4) correspond to assemblies of many polymer chains. Mean-field SCFT does not explicitly control if the structure contains hundreds or millions of chains. When modeling the chromosome, the system contains only one (or two, if homologous chromosomes are present) chain, which makes the SCFT prediction inaccurate.

More precisely, SCFT corresponds to a saddle-point evaluation of the polymer field theory. It replaces the functional integral over fluctuating auxiliary fields with its dominant field. As a consequence, SCFT neglects fluctuations around the saddle point of the chemical potential fields. This approximation is most accurate when the fluctuation is weak, such as in a dense polymer melt. However, in single-chain settings, fluctuation can become non-negligible. For future research, the model needs to go beyond mean-field to incorporate more biophysically realistic interactions and their fluctuation fields.

6.3. Conclusion

In this project, we implemented codebases of an SCFT solver and a workflow to infer polymer block sequences from Hi-C data. Using these tools, we demonstrated how thermodynamic parameter settings in SCFT affect the phase and free-energy landscape by comparing density-derived overlap/contact maps with experimental Hi-C contact patterns.

Several important steps remain before this framework can be used to make biologically grounded predictions. Although we showed how interaction parameters can be tuned to improve agreement with Hi-C data, other key model parameters (e.g., the degree of polymerization N and system size L) were chosen heuristically and are not yet constrained by independent experimental measurements. Establishing such a mapping is challenging not only because of the limited time scope of this project, but also because the current simulations are restricted to relatively short genomic segments (2 Mb and 12 Mb) and rely on a mean-field description that neglects field fluctuations.

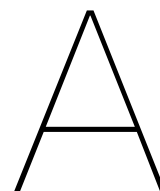
Despite these limitations, our results indicate that polymer field-based approaches have significant potential for chromosome modeling. In particular, the free-energy landscape provides an additional perspective on chromatin organization by linking structural morphologies and entropy to thermodynamic driving forces. Finally, the publicly available, GPU-accelerated code developed in this thesis (https://github.com/yuchenyan0107/SCFT_DNA) provides a reproducible starting point for further methodological development and for broader biophysics applications involving multiblock polymer self-assembly.

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Supplementary information

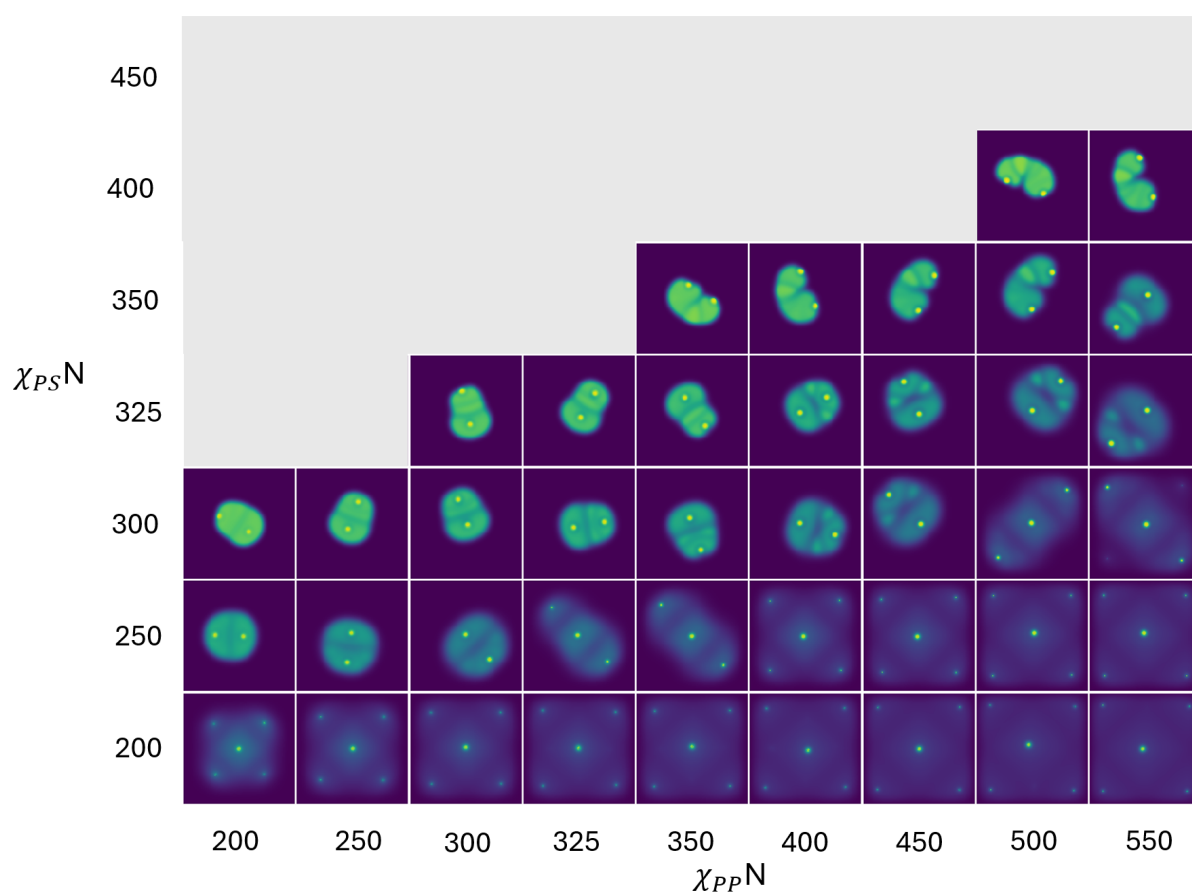


Figure A.1: The polymer density maps of the simulated results from the parameter sweep of $\chi_{ps}N$ and $\chi_{pp}N$.

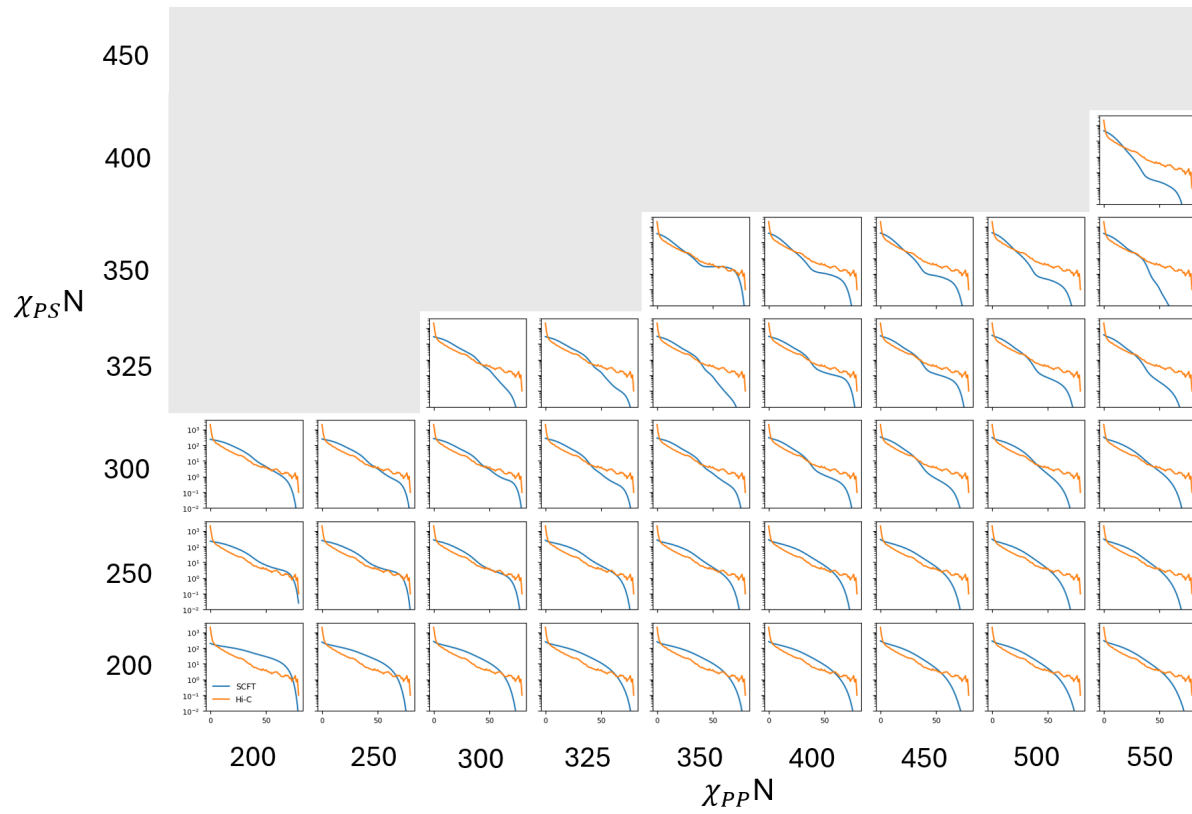


Figure A.2: The decay profiles of the simulated results from the parameter sweep of $\chi_{ps}N$ and $\chi_{pp}N$.

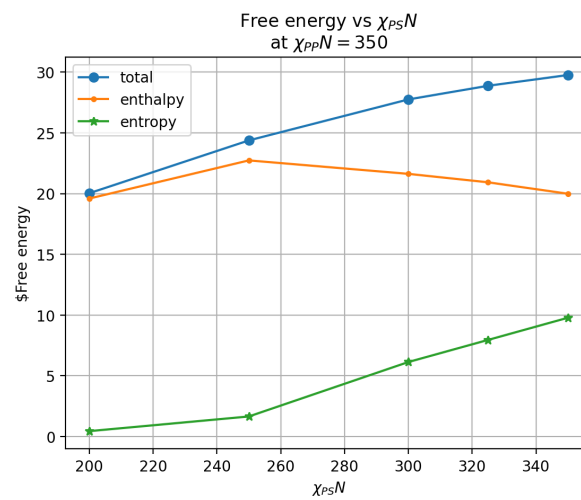


Figure A.3: Enthalpic, entropic, and total free energy versus $\chi_{ps}N$, at $\chi_{pp}N = 350$.

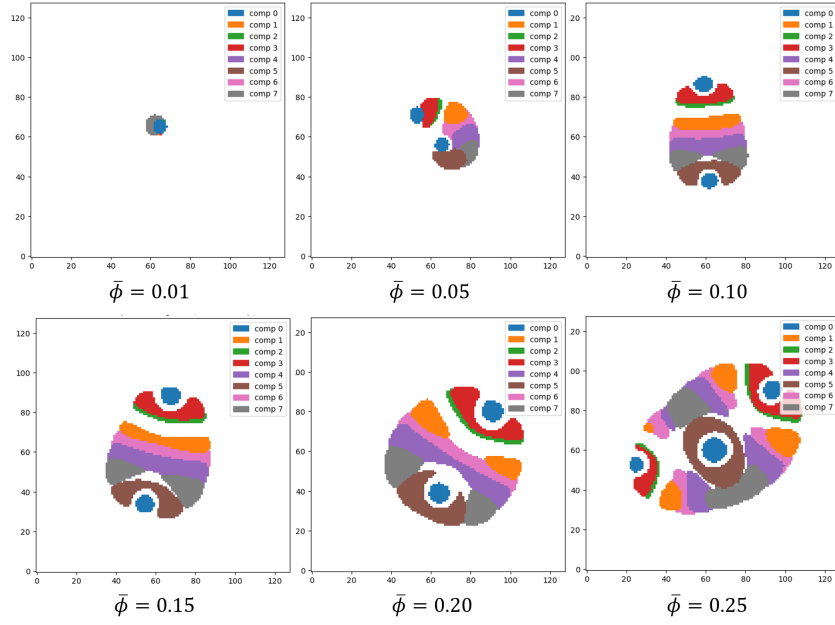


Figure A.4: The polymer density maps of the simulated results from the parameter sweep of $\bar{\phi}$ from 0.01 to 0.25. The box size $L = 30$.

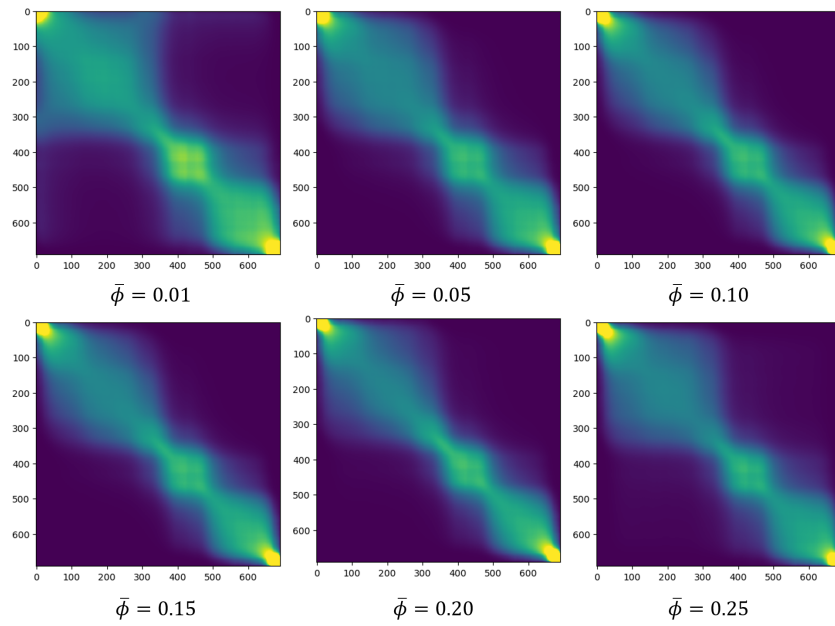


Figure A.5: The polymer density maps of the simulated results from the parameter sweep of $\bar{\phi}$ from 0.01 to 0.25. The box size $L = 30$.

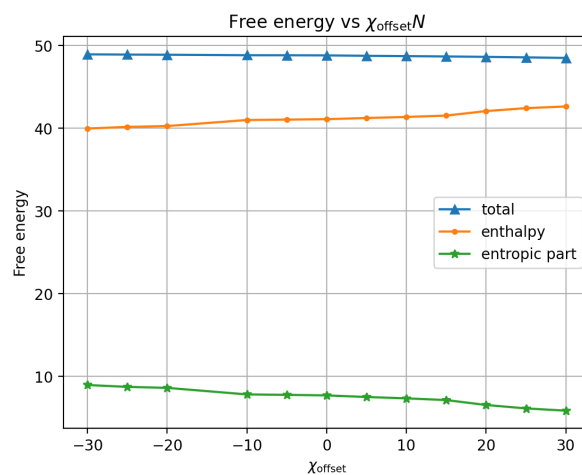


Figure A.6: Enthalpic, entropic, and total free energy versus $\chi_{\text{offset}}N$, corresponding to Figure 5.8.

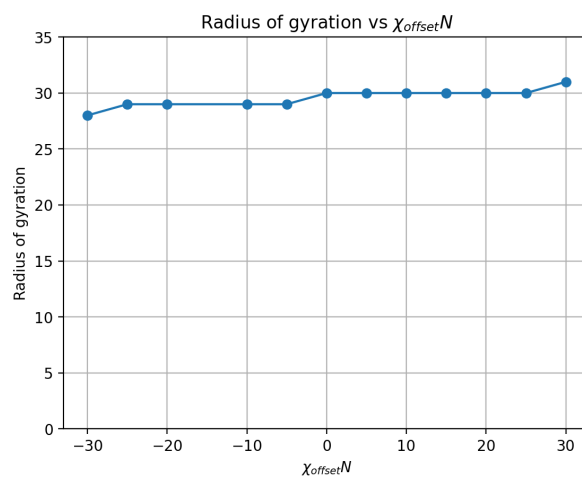


Figure A.7: The radius of gyration as a function of χ_{offset} .

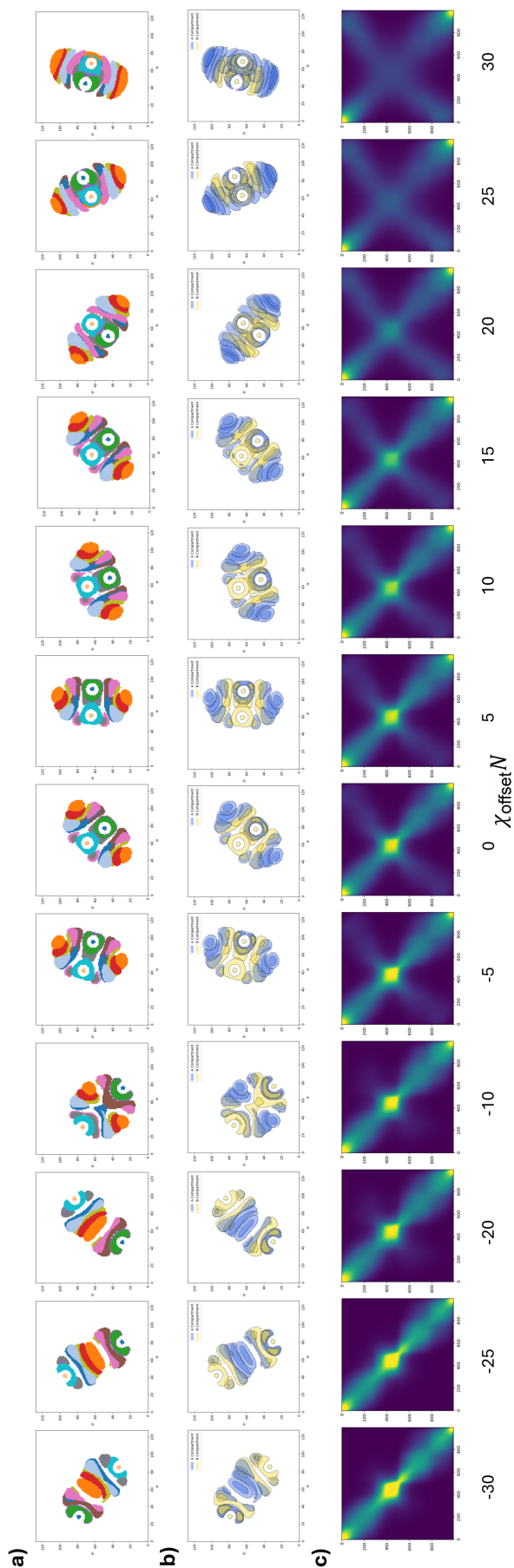


Figure A.8: **(a)** Polymer density distributions for the parameter sweep, colored by inferred polymer block type. **(b)** The same conformations with blocks grouped by compartment annotation (A: blue vs B: yellow). **(c)** Contact maps computed from the corresponding polymer conformations for each $\chi_{\text{offset}}N$ condition.